

Submersible Motor Pump

Amarex KRT

50 Hz

Type Series Booklet



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Type Series Booklet Amarex KRT

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Waste Water

Submersible Motor Pump

Amarex KRT



Main applications

- Waste water management
- Service water supply systems
- Disposal
- Waste water treatment plants
- Sludge disposal

Fluids handled

- Waste water with faeces
- Activated sludge
- Digested sludge
- Raw sludge
- Gas-containing liquids
- Industrial waste water

Operating data

Table 1: Operating properties

Characteristic		Value
Flow rate	Q [m³/h]	≤ 10080
	Q [l/s]	≤ 2800
Head	H [m]	≤ 120
Fluid temperature	T [°C]	≤ +60
Motor rating	P ₂ [kW]	0,8 - 850

Design details

Design

- Fully floodable submersible motor pump
- Not self-priming
- Close-coupled design

Drive

- Three-phase asynchronous squirrel-cage motor
- Type of protection Ex db IIB (applies to explosion-proof pump sets only)
- Enclosure: IP68 to EN 60529/IEC529

Shaft seal

Depending on the motor design:

- Two bi-directional mechanical seals in tandem arrangement with liquid reservoir, with leakage chamber on some models
- Double cartridge seal with leakage chamber

Impeller type

- Various application-oriented impeller types (⇒ Page 14)

Bearings

- Various application-oriented bearings (⇒ Page 8)

Drive end:

- Grease-packed bearings sealed for life
- Maintenance-free

Pump end:

- Standard bearings
 - Grease-packed bearings sealed for life
 - Maintenance-free
- Reinforced bearings
 - Grease-lubricated bearings
 - Can be re-lubricated

Designation

Example: Amarex KRT K 150-503/155 4 UN G-D IE3

Table 2: Designation key

Code	Description
Amarex KRT	Type series
K	Impeller type
	S/S-max Impeller with cutter
	F/F-max Vortex impeller
	E/E-max Closed single-channel impeller
	D/D-max Open, diagonal single-vane impeller / open, radial multi-vane impeller
	K/K-max Closed multi-channel impeller
150	Nominal discharge nozzle diameter [mm]
503	Maximum nominal impeller diameter [mm]
155	Motor size
4	Number of motor poles
UN	Motor version (⇒ Page 8)
	UN/UE Non-explosion-proof, for fluid temperatures of up to 40 °C ¹⁾
	UF Non-explosion-proof, for fluid temperatures of up to 40 °C, ambient temperatures of up to 55 °C
	WN/WE Non-explosion-proof, for fluid temperatures of up to 60 °C ¹⁾
	XN/XE Explosion protection Ⓢ II2G Ex db h IIB T3 Gb, for fluid temperatures of up to 40 °C ¹⁾
	XF Explosion protection Ⓢ II2G Ex db h IIB T3 Gb, for fluid temperatures of up to 40 °C, ambient temperatures of up to 55 °C
	YN/YE Explosion protection Ⓢ II2G Ex db h IIB T4 Gb, for fluid temperatures of up to 40 °C ¹⁾
	ZN/ZE Explosion protection Ⓢ II2G Ex db h IIB T3 Gb, for fluid temperatures of up to 60 °C ¹⁾
G	Material variant (⇒ Page 6)
	G Standard variant, grey cast iron
	G1 Like G, impeller made of duplex stainless steel
	G2 Like G, impeller made of white cast iron
	GH Like G, impeller and discharge cover made of white cast iron
	H Wetted components made of white cast iron
	C1 Wetted parts made of duplex stainless steel, mechanical seal with elastomer bellows, bolts/screws made of A4
	C2 Wetted parts made of duplex stainless steel, mechanical seal with covered spring, bolts/screws made of 1.4462
D	Installation type (⇒ Page 33)
	D Stationary dry installation, vertical (S1 duty)
	H Stationary dry installation, horizontal (S1 duty)
	K Stationary wet installation (S1 duty with motor outside of the fluid possible) with guide wire arrangement or guide rail arrangement
	S Stationary wet installation (S1 duty with submerged motor) with guide wire arrangement or guide rail arrangement
	P Wet installation of transportable model (S1 duty with submerged motor)
IE3	Motor efficiency classification ²⁾
	³⁾ No efficiency classification
	IE3 Premium Efficiency

¹⁾ Maximum fluid temperature and ambient temperature

²⁾ The IEC 60034-30 standard is not binding for submersible motor pumps. Efficiencies are calculated / determined by analogy with the measurement method specified in IEC 60034-2. The marking is used for submersible motors that achieve efficiency levels similar to those of standardised motors acc. to the IEC 60034-30 standard.3759

³⁾ Blank

Materials

Table 3: Overview of available materials

Part No.	Description	Material variant						
		G	G1	G2	GH	H	C1	C2
Pump set								
101	Pump casing	EN-GJL-250			EN-GJN-HB555		1.4517	
135	Wear plate for D/D-max impeller	EN-GJL-250 ⁴⁾		EN-GJN-HB555		-		
163	Discharge cover	EN-GJL-250			EN-GJN-HB555		1.4517	
210	Shaft	1.4021 (⇒ Page 8)					1.4021/1.4462/C45+N (⇒ Page 8)	
230	Impeller ⁵⁾	EN-GJL-250	JL 1.4517	EN-GJN-HB555			1.4517	
350	Bearing housing	EN-GJL-250					1.4517/EN-GJL-250	
412	O-ring	Nitrile butadiene rubber (NBR)						Viton (FKM)
433.01 / 433	Mechanical seal (drive end) Double mechanical seal (drive end)	Carbon/SiC						
433.02 / 433	Mechanical seal (pump end) Double mechanical seal (pump end)	SiC / SiC						
502	Casing wear ring ⁶⁾	EN-GJL-250			VG 434			
66- 2	Cooling jacket	1.4571			-			
811	Motor housing	EN-GJL-250					1.4517	
824	Power cable	(⇒ Table 7)						
900	Bolts/screws	A4 ⁷⁾						1.4462
Installation parts								
572	Guide wire suspension bracket	1.4571 up to DN 200; EN-GJL-250 from size K200-502						1.4571
59-24	Guide wire	1.4401						1.4401/ Tefzel
72-1	Flanged bend	EN-GJL-250			EN-GJN-HB555		1.4517	
732	Claw	EN-GJL-250 or EN-GJS-400-15/EN-GJS-500-7					1.4517	
885	Lifting chain / lifting rope	Lifting chain: 1.4404 Lifting rope: polyamide / polypropylene					Lifting rope: polypropylene	
892	Foot plate / feet	1.0038 + Z					1.4571	1.4517/ 1.4462
894	Mounting bracket	1.4571 up to DN 200; 1.0038 + Z from size K200-502						1.4571

Description of materials

EN-GJL-250 grey cast iron (lamellar graphite cast iron)

Lamellar graphite cast iron to EN 1561 is the most widely used cast material for handling municipal sewage, waste water and sludges as well as stormwater and surface water. It is suitable for neutral fluids which are only slightly aggressive and cause little wear. The pH should be ≥ 6.5 , the sand content ≤ 0.5 g/l.

Nodular cast iron EN-GJS-400-15

Its ductile structure, mechanical properties and wear resistance make this nodular cast iron to EN 1561 a suitable propeller material that is also suitable for handling digestion substrate.

Duplex stainless steel (1.4517 or technically equivalent material)

This type of carbon steel is resistant to cavitation, has excellent strength values and is used for high circumferential speeds. An excellent resistance to pitting corrosion makes ferritic-austenitic stainless carbon steel a popular choice for pumping acidic waste water with a high chloride content as well as seawater and brackish water. Thanks to its good chemical resistance, e.g. against waste water containing phosphorous and sulphuric acid, this material is used in a wide range of applications in the chemical industry and process engineering. Pumps made of duplex stainless steel have a very long service life, even when handling brines, chemical waste water (pH 1 - 12), grey water and landfill leachate.

Wear-resistant white cast iron (EN-GJN-HB555 [XCR14] or technically equivalent material)

Wear-resistant white cast iron is suitable for handling highly abrasive fluids containing sand, ash or iron ore sinter, for example. It has a Rockwell hardness of at least 54 HRC, which is higher than that of hardened chrome steel. Owing to its hardness, the chromium-molybdenum alloy cast iron features a notably higher wear resistance than EN-GJL-250 grey cast iron and other cast materials. The pH should be ≥ 6.5 .

⁴ D impeller: EN-GJL-250; D-max impeller: EN-GJS-400-15

⁵ D impeller: EN-GJL-250, with hardened edges; D-max impeller: EN-GJS-400-15, with hardened edges

⁶ For E impeller and K impeller

⁷ Corresponds to 1.4571

Product benefits

- Absolutely water-tight resin-sealed cable entries prevent any water from entering the motor – even in the event of a damaged cable.
- Reliable operation ensured by leakage sensors signalling any ingress of moisture into the motor
- Reliable operation ensured by sensors monitoring the motor temperature and preventing overheating
- Non-clogging low-maintenance design with large free passages reduces clogging risk and, consequently, maintenance work.
- High-efficiency motors and variable hydraulic systems for optimum hydraulic efficiency and energy efficiency

Material variants C1 and C2:

- Long service life due to corrosion-resistant wetted parts made of stainless steel

Product information

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see <https://www.ksb.com/en-global/company/corporate-responsibility/reach>.

Acceptance tests and warranty

Functional test

- Every pump undergoes functional testing to KSB standard ZN 56525.

K/K-max impeller:

- Operating data is guaranteed to DIN EN ISO 9906/2B. It can also be guaranteed to DIN EN ISO 9906/1B, 3B and 1U (with re-evaluation of D2 and η).

S, F/F-max, E/E-max, D/D-max impellers:

- Operating data is guaranteed to DIN EN ISO 9906/3B. It can also be guaranteed to DIN EN ISO 9906/2B.

Power input P₂ of the pump below 10 kW.

- Operating data is guaranteed to DIN EN ISO 9906 §4.4.2. Other acceptance grades are not approved.

Acceptance inspections/tests

- Acceptance test to ISO/DIN or comparable standards available against a surcharge.

Warranty

- Quality is assured by means of an audited and certified quality assurance system to DIN EN ISO 9001.

Selection information

- The indicated heads and performance data apply to material variant G, for fluids with a density $\rho = 1 \text{ kg/dm}^3$ and a kinematic viscosity $\nu \leq 20 \text{ mm}^2/\text{s}$.

Impeller type

- S, F, E, and D impellers can only be supplied with the documented impeller diameters. Indicate the pump set designation and the impeller diameter in the purchase order.

- Impellers K and D-max are trimmed to the duty point. Indicate the H/Q data or the impeller diameter in the purchase order. In the hydraulic selection program, the impeller diameter is automatically computed based on the H/Q data and added to the designation of the pump set.

Pump input power

- Adjust the power input to the density of the fluid handled:
 $P_2 \text{ (required)} = \rho \text{ [kg/dm}^3\text{]} \text{ (fluid handled)} \times P_2 \text{ (documented)}$
- Select the operating point with the largest power input within an operating range. Select a motor size providing a power reserve to compensate the tolerances in the system characteristic / pump characteristic.

Table 4: Recommended motor power reserve⁸⁾

P ₂ [kW]	Reserve	
	Mains operation	With frequency inverter
≤ 30	10 %	15 %
> 30	5 %	10 %

- For installation types D and K (with cooling jacket) with motors UN/XN/YN an additional power reserve of 1.5 kW must always be added for the cooling circuit.

i In the case of waste water, too low a flow velocity in the discharge line will lead to clogging and increased wear. The minimum flow velocity required in vertical risers was determined experimentally as a function of the pipe diameter used and shown to be between 1.2 m/s for nominal size 100 and 2.4 m/s for nominal size 1000. For more information on flow velocities required in horizontal and vertical discharge lines refer to "KSB Know-how: Planning Information for Amarex KRT Submersible Motor Pumps".

i In the case of waste water, too low a circumferential speed of the impeller will lead to clogging of the hydraulic system (operation on a frequency inverter). A minimum circumferential speed (measured at the impeller diameter) of 12 m/s must be observed.⁹⁾

⁸ If larger power reserves are stipulated by local regulations, these larger reserves must be provided.

⁹ For F impellers, a circumferential speed below 12 m/s is permissible.

Overview of product features / selection tables

Overview of product features

Table 5: Material variants G, G1, G2, GH

Feature	Motor							
2 poles	3 2 E ... 26 2 E	30 2 E ... 37 2 E	55 2 E ... 75 2 E	18 2 F ... 75 2 F	-	-	-	-
4 poles	2 4 E ... 22 4 E	30 4 E ... 37 4 E	45 4 E ... 75 4 E	15 4 F ... 75 4 F	80 4 N ... 110 4 N	130 4 N ... 175 4 N	200 4 N ... 350 4 N	-
6 poles	7 6 E ... 18 6 E	22 6 E ... 30 6 E	31 6 E ... 55 6 E	15 6 F ... 55 6 F	60 6 N ... 100 6 N	120 6 N ... 165 6 N	190 6 N ... 480 6 N	530 6 N ... 850 6 N
8 poles	-	11 8 E ... 22 8 E	30 8 E ... 45 8 E	15 8 F ... 45 8 F	50 8 N ... 75 8 N	90 8 N ... 130 8 N	150 8 N ... 400 8 N	460 8 N ... 760 8 N
10 poles	-	-	-	-	-	40 10 N ... 90 10 N	110 10 N ... 350 10 N	390 10 N ... 660 10 N
12 poles	-	-	-	-	-	-	105 12 N ... 300 12 N	340 12 N ... 560 12 N
Shaft material								
Shaft	1.4021							
Bearing assembly	Grease-packed rolling element bearings sealed for life ¹⁰⁾				Regreasable rolling element bearings (pump end) Grease-packed rolling element bearings sealed for life (drive end)			
Explosion protection								
Version U	Non-explosion-proof							
Version X	⊕ II2G Ex db h IIB T3 Gb							-
Version Y	⊕ II2G Ex db h IIB T4 Gb			-	⊕ II2G Ex db h IIB T4 Gb		-	-
Version W	Non-explosion-proof			-	Non-explosion-proof			-
				-				
Version Z	⊕ II2G Ex db h IIB T3 Gb			-	⊕ II2G Ex db h IIB T3 Gb			-
Motor								
Starting method	DOL / star-delta (690 V only DOL) ¹¹⁾							DOL
Voltage	400 V / 380 V ¹²⁾ / 415 V ¹²⁾ / 500 V ¹²⁾ / 690 V ¹²⁾							400 V / 690 V ¹²⁾
Cooling	Cooled by surrounding fluid / air cooling ¹³⁾		With cooling jacket	Cooled by surrounding fluid / with cooling jacket				
Immersion depth	≤ 30 m							
Power cable								
Type	See "Overview of power cables" (⇒ Table 7)							
Length	10 m / ≤ 40 m ¹²⁾							
Cable entry	Absolutely water-tight							
Sealing elements								
Elastomer seals	Nitrile butadiene rubber NBR / Viton = fluorocarbon rubber FPM ¹²⁾							

¹⁰⁾ For versions with D impeller and motors 55 2 E/F ... 75 2 E/F, 45 4 E/F ... 75 4 E/F, 31 6 E/F ... 55 6 E/F, 30 8 E/F ... 45 8 E/F: regreasable rolling element bearings (pump end) / grease-packed rolling element bearings sealed for life (drive end)

¹¹⁾ Depending on the motor size and voltage

¹²⁾ Optional

¹³⁾ Optional for motors 11 2 E ... 26 2 E, 7 4 E ... 22 4 E, 7 6 E ... 18 6 E

Feature	Motor							
2 poles	3 2 E ... 26 2 E	30 2 E ... 37 2 E	55 2 E ... 75 2 E	18 2 F ... 75 2 F	-	-	-	-
4 poles	2 4 E ... 22 4 E	30 4 E ... 37 4 E	45 4 E ... 75 4 E	15 4 F ... 75 4 F	80 4 N ... 110 4 N	130 4 N ... 175 4 N	200 4 N ... 350 4 N	-
6 poles	7 6 E ... 18 6 E	22 6 E ... 30 6 E	31 6 E ... 55 6 E	15 6 F ... 55 6 F	60 6 N ... 100 6 N	120 6 N ... 165 6 N	190 6 N ... 480 6 N	530 6 N ... 850 6 N
8 poles	-	11 8 E ... 22 8 E	30 8 E ... 45 8 E	15 8 F ... 45 8 F	50 8 N ... 75 8 N	90 8 N ... 130 8 N	150 8 N ... 400 8 N	460 8 N ... 760 8 N
10 poles	-	-	-	-	-	40 10 N ... 90 10 N	110 10 N ... 350 10 N	390 10 N ... 660 10 N
12 poles	-	-	-	-	-	-	105 12 N ... 300 12 N	340 12 N ... 560 12 N
Shaft seal	Mechanical seal with elastomer bellows / double cartridge mechanical seal ¹²⁾			Double cartridge mechanical seal	Mechanical seal with elastomer bellows / pump-end mechanical seal with covered spring ¹²⁾			Stationary mechanical seal with covered spring
Monitoring equipment								
Winding temperature, versions U, W; installation types S, P	Temperature switches (bimetal) in the winding			-	Temperature switches (bimetal) in the winding			
Winding temperature, versions X, Z; installation types S, P	Temperature switches (bimetal) in the winding, plus PTC thermistors for explosion protection Temperature switches (bimetal) in the winding, plus temperature switches (bimetal) for explosion protection ¹⁴⁾			-	Temperature switches (bimetal) in the winding, plus PTC thermistors for explosion protection			-
Winding temperature; installation types D, H, K	PTC thermistor	-		PTC thermistor / Pt100 resistance thermometer ¹⁵⁾	PTC thermistor			
Coolant temperature; installation types D, K	-			-	PTC thermistor			
Bearing temperature	-		Pt100 resistance thermometer (pump end) ¹²⁾	Pt100 resistance thermometer (pump end) ¹⁶⁾ Pt100 resistance thermometer (drive end) ¹⁵⁾	Pt100 resistance thermometer (pump end) Pt100 resistance thermometer (drive end) ¹²⁾			Pt100 resistance thermometer (pump end) Pt100 resistance thermometer (drive end)
Leakage in the motor space	Leakage sensor in the motor space							
Mechanical seal leakage	-			Float switch in leakage area				
Vibration sensor	-			Internal vibration sensor ¹⁵⁾	Internal vibration sensor ¹²⁾			
Coating	Environmentally friendly KSB standard coating (colour RAL 5002) / 250 µm, 2-component epoxy coating ¹²⁾							
Maximum fluid temperature and ambient temperature								

¹⁴⁾ Only for motors 3 2 E, 2 4 E, 3 4 E

¹⁵⁾ Option: Premium package

¹⁶⁾ Option: Basic Plus package or Premium package

Feature	Motor							
2 poles	3 2 E ... 26 2 E	30 2 E ... 37 2 E	55 2 E ... 75 2 E	18 2 F ... 75 2 F	-	-	-	-
4 poles	2 4 E ... 22 4 E	30 4 E ... 37 4 E	45 4 E ... 75 4 E	15 4 F ... 75 4 F	80 4 N ... 110 4 N	130 4 N ... 175 4 N	200 4 N ... 350 4 N	-
6 poles	7 6 E ... 18 6 E	22 6 E ... 30 6 E	31 6 E ... 55 6 E	15 6 F ... 55 6 F	60 6 N ... 100 6 N	120 6 N ... 165 6 N	190 6 N ... 480 6 N	530 6 N ... 850 6 N
8 poles	-	11 8 E ... 22 8 E	30 8 E ... 45 8 E	15 8 F ... 45 8 F	50 8 N ... 75 8 N	90 8 N ... 130 8 N	150 8 N ... 400 8 N	460 8 N ... 760 8 N
10 poles	-	-	-	-	-	40 10 N ... 90 10 N	110 10 N ... 350 10 N	390 10 N ... 660 10 N
12 poles	-	-	-	-	-	-	105 12 N ... 300 12 N	340 12 N ... 560 12 N
Version U	40 °C ^{17), 18)}			40 °C ¹⁷⁾ and 55 °C ¹⁸⁾	40 °C ^{17), 18)}			
Version X	40 °C ^{17), 18)}			40 °C ¹⁷⁾ and 55 °C ¹⁸⁾	40 °C ^{17), 18)}			-
Version Y	40 °C ^{17), 18)}			-	40 °C ^{17), 18)}			-
Versions W, Z	60 °C ^{17), 18)}			-	60 °C ^{17), 18)}			-
Tests/inspections								
Hydraulic system	KSB standard (ZN 56525) / S impeller, D impeller, E impeller, F impeller (ISO 9906/A) ¹²⁾ / K impeller (ISO 9906//1/2/A) ¹²⁾							
General	KSB standard (ZN 56525)							
Installation type								
Stationary, with guide wire arrangement	Installation depth 4.5 m / 15 m ¹⁹⁾ / ≤ 30 m ¹²⁾							
Transportable	Up to size 300-403 (except sizes 200-503/504/505, 200-632)			-	Up to size 300-403 (except sizes 200-503/504/505, 200-632)			-
Stationary, with guide rail arrangement	Installation depth 4.5 m / ≤ 30 m ¹²⁾							
Stationary, dry-in-stalled	- air cooling ¹³⁾			With cooling jacket	With cooling jacket			
Stationary, dry-in-stalled, horizontal, with maintenance aid	-			With Amaslide ¹²⁾	-			

¹⁷⁾ Fluid temperature

¹⁸⁾ Ambient temperature

¹⁹⁾ From size K200-503

Table 6: Material variants H, C1, C2

Feature	Motor						
2 poles	3 2 E ... 26 2 E	30 2 E ... 37 2 E	55 2 E ... 75 2 E	-	-	-	-
4 poles	2 4 E ... 22 4 E	30 4 E ... 37 4 E	45 4 E ... 75 4 E	80 4 N ... 110 4 N	130 4 N ... 175 4 N	200 4 N ... 350 4 N	-
6 poles	7 6 E ... 18 6 E	22 6 E ... 30 6 E	31 6 E ... 55 6 E	60 6 N ... 100 6 N	120 6 N ... 165 6 N	190 6 N ... 480 6 N	530 6 N ... 850 6 N
8 poles	-	11 8 E ... 22 8 E	30 8 E ... 45 8 E	50 8 N ... 75 8 N	90 8 N ... 130 8 N	150 8 N ... 400 8 N	460 8 N ... 760 8 N
10 poles	-	-	-	-	40 10 N ... 90 10 N	110 10 N ... 350 10 N	390 10 N ... 660 10 N
12 poles	-	-	-	-	-	105 12 N ... 300 12 N	340 12 N ... 560 12 N
Shaft material for material variant H							
Shaft	1.4021						
Shaft material for material variants C1, C2							
Shaft	1.4462 / C45+N			1.4021 ²⁰⁾	1.4021		
Suction flange	Drilled to DN 2501 ¹²⁾						
Bearing assembly	Grease-packed rolling element bearings sealed for life			Regreasable rolling element bearings (pump end) Grease-packed rolling element bearings sealed for life (drive end)			
Explosion protection							
Version U	Non-explosion-proof						
Version X	⊗ II2G Ex db h IIB T3 Gb						-
Version Y	⊗ II2G Ex db h IIB T4 Gb					-	-
Version W	Non-explosion-proof						
Version Z	⊗ II2G Ex db h IIB T3 Gb						-
Motor							
Starting method	DOL / star-delta (690 V only DOL) ¹¹⁾						DOL
Voltage	400 V / 380 V ¹²⁾ / 415 V ¹²⁾ / 500 V ¹²⁾ / 690 V ¹²⁾						400 V / 690 V ¹²⁾
Cooling	Cooled by surrounding fluid						
Immersion depth	≤ 30 m						
Power cable							
Type	See "Overview of power cables" (⇒ Table 7)						
Length	10 m / ≤ 40 m ¹²⁾						
Cable entry	Absolutely water-tight						
Sealing elements							
Elastomer seals	Nitrile butadiene rubber NBR / Viton = fluorocarbon rubber FPM ¹²⁾ / fluorocarbon rubber FPM (C2)						
Shaft seal	C1: Mechanical seal with elastomer bellows ²¹⁾ H, C2: Double cartridge mechanical seal			C1: Mechanical seal with elastomer bellows ²²⁾ H, C2: Pump-end mechanical seal with covered spring		Stationary mechanical seal with covered spring	
Monitoring equipment							

²⁰⁾ For maximum nominal impeller diameters 400 mm / 401 mm / 402 mm / 403 mm: 1.4462

²¹⁾ Optional: Double cartridge mechanical seal

²²⁾ Optional: Mechanical seal with covered spring

Feature	Motor						
2 poles	3 2 E ... 26 2 E	30 2 E ... 37 2 E	55 2 E ... 75 2 E	-	-	-	-
4 poles	2 4 E ... 22 4 E	30 4 E ... 37 4 E	45 4 E ... 75 4 E	80 4 N ... 110 4 N	130 4 N ... 175 4 N	200 4 N ... 350 4 N	-
6 poles	7 6 E ... 18 6 E	22 6 E ... 30 6 E	31 6 E ... 55 6 E	60 6 N ... 100 6 N	120 6 N ... 165 6 N	190 6 N ... 480 6 N	530 6 N ... 850 6 N
8 poles	-	11 8 E ... 22 8 E	30 8 E ... 45 8 E	50 8 N ... 75 8 N	90 8 N ... 130 8 N	150 8 N ... 400 8 N	460 8 N ... 760 8 N
10 poles	-	-	-	-	40 10 N ... 90 10 N	110 10 N ... 350 10 N	390 10 N ... 660 10 N
12 poles	-	-	-	-	-	105 12 N ... 300 12 N	340 12 N ... 560 12 N
Winding temperature, versions U, W	Temperature switches (bimetal) in the winding						
Winding temperature, versions X, Y	Temperature switches (bimetal) in the winding, plus PTC thermistors for explosion protection Temperature switches (bimetal) in the winding, plus temperature switches (bimetal) for explosion protection ¹⁴⁾			Temperature switches (bimetal) in the winding, plus PTC thermistors for explosion protection			
Bearing temperature	-		Pt100 resistance thermometer (pump end) ¹²⁾	Pt100 resistance thermometer (pump end) Pt100 resistance thermometer (drive end) ¹²⁾		Pt100 resistance thermometer (pump end) Pt100 resistance thermometer (drive end)	
Motor leakage	Leakage sensor in the motor space						
Coating	H: Environmentally friendly KSB standard coating (colour RAL 5002) / H: 250 µm, 2-component epoxy coating ¹²⁾ / C1, C2: Without coating						
Maximum fluid temperature and ambient temperature							
Version U	40 °C ^{17), 18)}			40 °C ^{23), 17), 18)} 30 °C ^{24), 17), 18)}			
Versions X, Y	40 °C ^{17), 18)}			40 °C ^{23), 17), 18)} 30 °C ^{24), 17), 18)}		-	
Versions W, Z	60 °C ^{17), 18)}						-
Tests/inspections							
Hydraulic system	KSB standard (ZN 56525) / S impeller, F impeller (ISO 9906/A) ¹²⁾ / K impeller (ISO 9906//1/2/A) ¹²⁾						
General	KSB standard (ZN 56525)						
Installation type							
Stationary, with guide wire arrangement	Installation depth 4.5 m / ≤ 30 m ¹²⁾						
Transportable	Installation depth 4.5 m				-		

²³ White cast iron H

²⁴ Stainless steel C

Table 7: Overview of power cables

Feature	S1BN8-F rubber-sheathed cable	S07RC4N8-F rubber-sheathed cable	TEHSITE Tefzel cable
Design	Standard	Optional	Optional
Rated voltage	1000 V	750 V	750 V
EMC screening	-	✓	-
Insulation material	EPR ²⁵⁾	EPR ²⁵⁾	ETFE ²⁶⁾
Max. continuous temperature of insulation	90 °C	90 °C	135 °C
For permanent immersion in waste water to DIN VDE 0282-16/HD22.16	✓	✓	✓

²⁵⁾ EPR = ethylene propylene rubber
²⁶⁾ ETFE = ethylene tetrafluoroethylene

Impellers

	Impeller with cutter (impeller type S/S-max)	Suitable for the following fluids: fluids containing coarse substances and/or long fibres
---	---	---

Further fluids (impeller type S/S-max):

- Domestic waste water
- Grey water
- Waste water with faeces

	Free-flow impeller (impeller type F/F-max)	Suitable for the following fluids: fluids containing solids and stringy material as well as fluids with entrapped air or entrapped gas
	Closed single-channel impeller (impeller type E/E-max)	Suitable for the following fluids: fluids containing solids and stringy material
	Open, diagonal single-channel impeller (impeller type D)	Suitable for the following fluids: fluids containing solid substances and long fibres
	Open radial multi-vane impeller (impeller type D-max)	Suitable for the following fluids: fluids containing solid substances and long fibres

Further fluids (impeller types F/F-max, E/E-max, D/D-max):

- Activated sludge
- Digested sludge
- Heating sludge
- Mixed water
- Raw waste water
- Raw sludge
- Recirculated sludge

	Closed multi-channel impeller (impeller type K/K-max)	Suitable for the following fluids: contaminated, solids-laden, non-gaseous fluids without stringy material
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Further fluids (impeller type K/K-max):

- Activated sludge
- Landfill waste water
- Industrial waste water
- Industrial grey water
- Mechanically treated waste water
- Pre-screened waste water
- Stormwater

Overview of fluids handled

The table below for your guidance is based on KSB's long-standing experience. The data are standard values and are not to be considered as generally binding recommendations. More detailed advice is available from KSB. Make use of our laboratory's expertise when selecting materials.

Table 8: Selection aid for materials and hydraulic systems per fluid

Fluid handled ²⁷⁾	Recommended material variant	Recommended impeller type ²⁸⁾	Comments, further recommendations
Grey water	G	K/K-max, D/D-max, E/E-max, F/F-max	Free passage > any solids contained, possibly pre-screened
River water	G	K/K-max, D/D-max, E/E-max, F/F-max	Free passage > any solids contained, possibly pre-screened
Stormwater	G	K/K-max, D/D-max, E/E-max, F/F-max	Free passage > any solids contained, possibly pre-screened
Waste water:			
▪ Untreated municipal waste water	G	F/F-max, S/S-max, D/D-max, E/E-max, K/K-max	ATV ²⁹⁾ recommends a free passage of 100 mm; minimum free passage: 76 mm
▪ Containing air and gas	G	F/F-max	Up to 8 %, contact KSB for fluids with high outgassing rates
Sludges:			
▪ Raw sludge	G	F/F-max, D/D-max, E/E-max	Pumpable up to a dry substance content of: 13 % (D), 8 % (F/D-max), 6 % (E)
▪ Digested sludge	G	F/F-max, D/D-max, E/E-max	Pumpable up to a dry substance content of: 13 % (D), 8 % (F/D-max), 6 % (E)
▪ Activated sludge	G	D/D-max, K/K-max	Pumpable up to a dry substance content of: 13 % (D), 8 % (D-max), 5 % (K)
Industrial waste water containing:			
▪ Paint suspensions	G	K/K-max	Solvent-free, observe the operator's instructions.
▪ Lacquer/paint/varnish suspensions	G	F/F-max, E/E-max	Solvent-free, contact KSB for silicone-free version.
▪ Fibrous material	G	F/F-max, S/S-max, D/D-max	-
▪ Chips/swarf	G2 / GH	K/K-max, F/F-max	Material variants G2 or GH, special mechanical seal, solids content < 5 g/l
▪ Abrasive substances ³⁰⁾	G2 / GH	K/K-max, F/F-max	Material variants G2 or GH, special mechanical seal, solids content < 5 g/l
Mildly acidic industrial waste water	C	K/K-max, F/F-max	pH ≥ 6.5, material variant C1 and FPM (Viton) O-rings
Neutral non-corrosive waste water:			
▪ Ammonium hydroxide	G	K/K-max	-
▪ Ammonium hydroxide 5 % NH ₄ OH	G	K/K-max	-
▪ Urea 25 % (NH ₂) ₂ -CO	G	K/K-max	-

²⁷⁾ For any fluids which are not listed in this table contact KSB.

²⁸⁾ The first impeller type listed should be given preference.

²⁹⁾ ATV = German regulatory body for waste water management

³⁰⁾ Severe hydroabrasive wear occurs if solids contents of approx. 0.5 g/l or higher are combined with circumferential speeds exceeding 20 m/s or low-flow conditions to the left of the duty point.

Fluid handled ²⁷⁾	Recommended material variant	Recommended impeller type ²⁸⁾	Comments, further recommendations
▪ Potassium hydroxide 10 % KOH	G	K/K-max	-
▪ Calcium hydroxide 5 % Ca(OH) ₂	G	K/K-max	-
▪ Sodium hydroxide 5 % NaOH	G	K/K-max	-
▪ Sodium carbonate 30 % Na ₂ CO ₃	G	K/K-max	-
Neutral, non-corrosive waste water containing:			
▪ Aliphatic hydrocarbons, e.g. oils, petrol, butane, methane	G	K/K-max	-
▪ Aromatic hydrocarbons, e.g. benzene, styrene	G	K/K-max	FPM (Viton) O-rings ³¹⁾
▪ Chlorinated hydrocarbons (e.g. tetrachloroethylene, ethylene chloride, chloroform, methylene chloride)	G	K/K-max	FPM (Viton) O-rings ³¹⁾
Highly abrasive industrial waste water causing wear (chemically neutral): ³²⁾			
▪ Water containing iron ore sinter	GH / H	K/K-max	Sinter content < 5 g/l: material variant GH Sinter content > 5 g/l: material variant H
▪ Lime milk containing quartz and pigment suspensions	GH / H	K/K-max	Lime milk content < 15 %: material variant GH Lime milk content > 15 %: material variant H
▪ Wash water containing solids	Wear-resistant white cast iron	K/K-max, F/F-max	Material selection based on fluid analysis
▪ Waste water containing dust or ash	GH / H	K/K-max	Material selection based on fluid analysis
Water/sand mixture	GH / H	K/K-max, F/F-max	Solids content < 5 g/l: material variant GH Solids content > 5 g/l: material variant H
Seawater	C	K/K-max, F/F-max	Material variant C2 ≤ 25 °C fluid temperature ³³⁾
Brackish water	G1 / C1	K/K-max, F/F-max	Material variant C1 or G1 (with 250 µm two-component epoxy resin), depending on salt content
Corrosive industrial waste water	C	K/K-max, F/F-max	Material variants C1 or C2, depending on fluid analysis

³¹⁾ The hydrocarbons mentioned may occur in very high concentrations due to the difference in specific weight and their low solubility. If this is the case, contact KSB.

³²⁾ The required material variants highly depend on the operating hours, rotational speed and flow velocity.

³³⁾ Higher fluid temperatures on request.

Shaft seal

Available shaft seal types per bearing bracket

Table 9: UE/WE/XE/YE/ZE motors

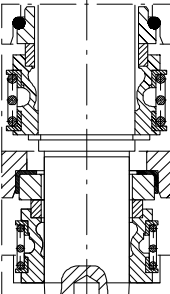
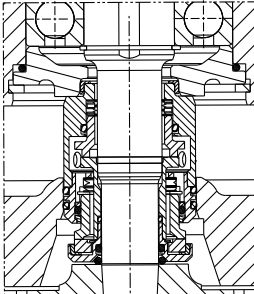
Mechanical seal with elastomer bellows ³⁴⁾	Double cartridge mechanical seal ³⁵⁾
	

Table 10: UF/XF motors

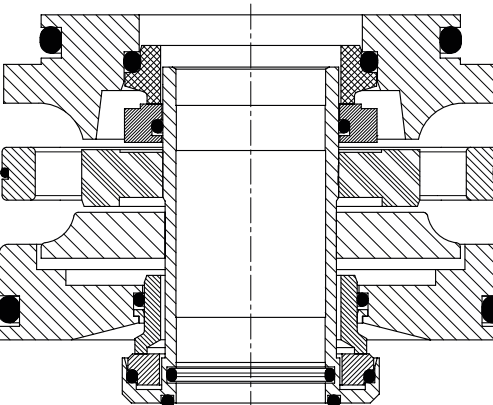
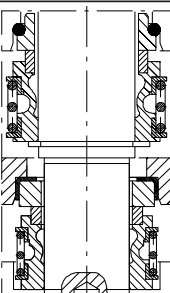
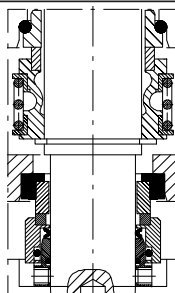
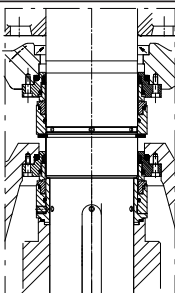
Double cartridge mechanical seal


Table 11: UN/WN/XN/YN/ZN motors

Mechanical seal with elastomer bellows ³⁴⁾	Pump-end mechanical seal with covered spring ³⁵⁾	Stationary mechanical seal with spring outside of fluid handled
		

For more information, see overview of product features. (⇒ Page 8)

³⁴⁾ For all types of waste water

³⁵⁾ For very abrasive fluids or fluids containing metallic particles (e.g. shavings from drilling)

Technical data
Table 12: Grey cast iron (G, G1, G2, GH)

Size	Impeller type	Material variant	Impeller				Installation type				Moment of inertia J ³⁶⁾
							D, H		K, S, P		
			Impeller channels	Free passage	Max. im-peller dia-meter	Min. im-peller dia-meter	Max. oper-ating pres-sure ³⁷⁾	Max. test pressure	Max. oper-ating pres-sure ³⁷⁾	Max. test pressure	
Qty	[mm]	[mm]	[mm]	[bar]	[bar]	[bar]	[bar]	[kgm ²]			
40-252	S	G	4	7	235	175	-	-	10	13	0,03
40-252	F	G, G1, G2, GH	-	25	210	150	10	15	7,6	9,8	0,03
40-252	K	G, G1, G2, GH	3	15	260	150	10	15	10	13	0,047
50-215	F	G, G1, G2, GH	-	42	210	130	10	15	8	10,5	0,019
50-216	S	G	4	4	210	175	-	-	9	12	0,018
50-216	F	G, G1, G2, GH	-	25	210	130	10	15	6,5	8,5	0,025
65-215	F	G, G1, G2, GH	-	65	210	130	6	9	5,5	7,5	0,025
65-216	E	G	1	65	180	140	6	9	5,5	7,5	0,02
65-217	F	G, G1, G2, GH	-	65	200	130	7	10,5	6	8	0,02
80-215	F	G, G1, G2, GH	-	76	200	130	5,5	8,5	4,5	6	0,025
80-216	F	G, G1, G2, GH	-	76	210	130	7	10,5	5	6,5	0,025
80-216	E	G	1	76	210	160	7	10,5	5,5	7,5	0,035
80-252	F	G, G1, G2, GH	-	76	265	150	6	9	6,3	8,2	0,14
80-253	F	G, G1, G2, GH	-	76	237	150	7,2	10,8	7,2	9,4	0,055
80-253	E	G	1	76	270	225	6	9	2,8	3,7	0,17
80-253	K	G, G1, G2, GH	2	33	220	140	6,6	10	6,6	8,6	0,15
80-315	D	G, G1	1	65	260	230	10	15	11	15	0,124
80-315	K	G, G1, G2	2	33	260	140	10	15	9,1	11,9	0,11
80-317	F	G, G1, G2, GH	-	76	240	150	10	15	7	9,1	0,14
80-317	D	G, G1	1	76	220	180	10	15	6	9	0,047
100-215	F	G, G1, G2, GH	-	100	210	130	6	9	4	5,5	0,025
100-253	E	G	1	76	270	210	5,5	8,5	4,5	6	0,15
100-253	D	G, G1	1	76	265	234	6	9	3,5	4,6	0,115
100-253	K	G, G1, G2, GH	2	76	256	200	4,5	7	3,5	4,6	0,15
100-254	F	G, G1, G2, GH	-	100	265	200	6	9	3,4	4,5	0,056
100-254	K	G, G1, G2, GH	2	71	256	210	6	9	2,5	3,2	0,07
100-315	F	G, G1, G2, GH	-	100	310	270	-	-	3,5	4,6	0,056
100-315	E	G	1	100	330	262	-	-	4,3	5,6	0,26
100-315	D	G, G1	1	75	222	196	10	15	6,8	8,8	0,065
100-315	K	G, G1, G2, GH	2	80	312	254	-	-	4	5,2	0,15
100-316	D	G, G1	1	85	306	270	10	15	3,6	4,7	0,233
100-316	F	G, G1, G2, GH	-	100	310	236	6	9	4,9	6,4	0,075
100-316	K	G, G1, G2, GH	2	76	309	235	6	9	4,9	6,4	0,13
100-317	E	G	1	76	328	286	7	10,5	6	8	0,25
100-400	K	G, G1, G2, GH	2	76	408	355	10	15	9,2	12	1,1
100-403	D	G, G1, G2	2	76	408	300	10	15	9,2	12	0,5
100-401	F	G, G1, G2, GH	-	100	390	325	10	13	7,6	9,8	0,248
100-401	E	G	1	80	412	389	-	-	5,1	6,6	0,6
100-401	K	G, G1, G2, GH	2	50	404	310	10	13	9,3	12,1	0,504
150-253	D	G, G1	1	100	254	225	6	9	1,9	2,4	0,15
150-315	F	G, G1, G2, GH	-	120	290	250	6	9	1,8	2,3	0,144
150-315	D	G, G1	1	100	317	280	6	9	3,3	4,3	0,289
150-317	D	G, G1, G2	2	76	309	250	6	9	5	6,5	0,17
150-317	E	G	1	110	320	254	6	9	3,1	4,1	0,31
150-317	K	G, G1, G2, GH	2	76	309	250	6	9	5	6,5	0,28
150-400	D	G, G1	1	100	363	326	10	15	5,2	6,8	0,573
150-400	K	G, G1, G2, GH	3	76	404	300	10	15	8,4	11	0,83
150-401	F	G, G1, G2, GH	-	135	390	270	10	15	4,2	5,5	0,248

³⁶ Data applies to maximum impeller diameter and impeller with water fill.

³⁷ Permissible operating pressure = inlet pressure + pressure at Q = 0

Size	Impeller type	Material variant	Impeller				Installation type				Moment of inertia ^{j36)}
							D, H		K, S, P		
			Impeller channels	Free passage	Max. impeller diameter	Min. impeller diameter	Max. operating pressure ³⁷⁾	Max. test pressure	Max. operating pressure ³⁷⁾	Max. test pressure	
			Qty	[mm]	[mm]	[mm]	[bar]	[bar]	[bar]	[bar]	
150-401	E	G	1	115	407	348	10	15	6,3	8,2	0,68
150-401	D	G, G1	1	110	384	370	10	15	5,3	6,9	0,999
150-403	D	G, G1, G2	2	76	408	340	10	15	8,5	11,1	0,53
150-403	K	G, G1, G2, GH	2	76	408	340	10	15	8,5	11,1	0,691
150-503	D	G, G1, G2	2	76	508	400	10	15	10	13	1,13
150-503	K	G, G1, G2	2	76	508	400	10	15	10	13	0,91
151-403	K	G, G1, G2, GH	2	76	408	340	10	15	9,3	11,9	0,691
200-401	E	G	1	120	400	319	10	15	5,7	7,4	0,86
200-315	D	G, G1	1	100	315	280	6	9	2,7	3,4	0,261
200-315	K	G, G1, G2, GH	3	70	295	245	6	9	1,9	2,4	0,22
200-316	K	G, G1, G2, GH	2	100	305	265	6	9	1,7	2,2	0,22
200-317	K	G, G1, G2, GH	3	76	309	240	3	4,5	3,5	5	0,4
200-318	K	G, G1, G2, GH	2	100	309	230	3	4,5	3	4	0,28
200-400	D	G, G1	1	100	375	355	10	15	4,2	5,5	0,825
200-402	D	G, G1, G2	3	80	408	300	10	15	6,5	9,8	0,5
200-402	K	G, G1, G2, GH	3	80	408	300	6,5	9,8	6,5	9,8	0,84
200-405	D	G, G1, G2	2	90	408	300	10	15	8,7	11,4	0,55
200-403	K	G, G1, G2, GH	2	90	408	300	6,5	9,8	6,5	9,8	0,931
200-504	K	G, G1	3	76	508	400	10	15	9,7	12,6	0,83
200-503	D	G, G1, G2	2	90	508	400	10	15	9,8	12,8	1,2
200-505	K	G, G1	2	90	508	400	10	15	9,8	12,8	1,636
250-400	D	G, G1	1	120	370	320	10	15	3,5	4,6	0,653
250-401	K	G, G1, G2, GH	2	105	400	310	10	15	6	7,8	0,55
250-402	D	G, G1, G2	3	106	398	300	10	15	6	7,8	0,6
250-403	K	G, G1, G2, GH	2	107	408	300	10	15	7	9,1	1,13
250-632	K	G, G1	3	105	638	500	10	15	10,6	13,8	5,684
250-900	K	G, G1	3	110	840	717	13	19,5	11,7	15,2	19,03
300-400	D	G, G1	1	150	408	375	10	15	1,7	2,2	0,925
300-400	K	G, G1, G2, GH	3	100	408	332	10	15	3,5	4,6	0,75
300-401	K	G, G1, G2, GH	2	135	408	367	10	15	2,3	2,9	0,75
300-402	D	G, G1, G2	3	100	408	300	10	15	3,5	4,6	0,63
300-403	K	G, G1, G2, GH	2	102	408	300	10	15	3,8	5	1,439
300-420	K	G, G1	3	100	408	370	6	9	5,6	7,3	0,95
300-500	K	G, G1	3	90	504	430	10	15	6,2	8	1,48
300-502	D	G, G1, G2	2	102	508	400	10	15	8	10,4	2,5
300-505	D	G, G1, G2	2	127	508	400	10	15	8,5	11	2,5
300-505	K	G, G1	2	102	508	400	10	15	8	10,4	2,919
350-500	K	G, G1	3	110	508	426	6	9	5,7	7,4	3,12
350-502	D	G, G1, G2	2	145	508	400	6	9	4,6	6	3,5
350-503	D	G, G1, G2	2	140	508	400	6	9	4,6	6	3,5
350-503	K	G, G1	2	140	508	400	6	9	4,6	6	4,073
350-632	K	G, G1	3	140	638	500	10	15	6,5	8,4	6,451
350-633	K	G, G1	2	135	638	500	10	15	9,4	12,2	6,979
350-710	K	G, G1	3	110	730	580	10	15	9,4	12,2	10,6
350-713	K	G, G1	2	125	738	580	13	19,5	12,2	16	14,557
400-500	K	G, G1	3	130	508	443	6	9	3,4	4,5	3,37
400-632	K	G, G1	3	142	638	527	6	9	5,7	7,41	9,074
400-900	K	G, G1	3	130	830	659	13	19,5	11,3	14,7	17,79
401-710	K	G, G1	3	165	739	587	10	15	8,8	11,5	16
401-713	K	G, G1	2	143	738	580	11,5	17,3	11,5	15	15,894
500-634	K	G, G1	3	132	638	448	4,5	6	4,5	6	10,248
501-710	K	G, G1	3	150	700	586	8,5	13	8,5	11,5	16
501-900	K	G, G1	3	178	908	721	9	13,5	8	10,3	45
600-520	K	G, G1	3	145	532	457	4	6	2,4	3,2	7,02

Size	Impeller type	Material variant	Impeller				Installation type				Moment of inertia J ⁽³⁶⁾
							D, H		K, S, P		
			Impeller channels	Free passage	Max. impeller diameter	Min. impeller diameter	Max. operating pressure ⁽³⁷⁾	Max. test pressure	Max. operating pressure ⁽³⁷⁾	Max. test pressure	
			Qty	[mm]	[mm]	[mm]	[bar]	[bar]	[bar]	[bar]	
600-710	K	G, G1	3	165	736	685	4	6	4,2	5,5	16,96
700-901	K	G, G1	3	180	908	760	9	13,5	7,2	9,3	50
700-902	K	G, G1	3	190	887	680	3,5	5	3,5	4,6	40
701-902	K	G, G1	3	190	887	680	-	-	5	6,5	40

Table 13: Industrial materials (H, C1, C2)

Size	Impeller type	Material variant	Impeller				Installation type		Moment of inertia J ⁽³⁶⁾
			Impeller channels	Free passage	Max. im-peller dia-meter	Min. im-peller dia-meter	S, P		
							Max. operating pressure ⁽³⁷⁾	Max. test pressure	
Qty	[mm]	[mm]	[mm]	[bar]	[bar]	[kgm ²]			
40-252	F	H, C1, C2	-	24	210	150	7,6	9,8	0,03
40-252	K	H, C1, C2	3	15	260	150	10	13	0,047
50-215	F	H, C1, C2	-	42	210	130	8	10,5	0,019
50-216	F	H, C1, C2	-	25	210	130	6,5	8,5	0,025
65-215	F	H, C1, C2	-	65	210	130	5,5	7,5	0,025
80-216	F	H, C1, C2	-	76	210	130	5	6,5	0,025
80-252	F	H, C1, C2	-	76	265	150	6,3	8,2	0,14
80-253	F	H, C1, C2	-	76	237	150	7,2	9,4	0,055
80-253	K	H, C1, C2	2	33	220	140	7,2	9,4	0,15
100-253	K	H, C1, C2	2	76	256	200	3,5	5	0,15
100-254	F	H, C1, C2	-	100	265	200	3,4	4,5	0,056
100-254	K	H, C1, C2	2	71	256	210	2,5	3,2	0,07
100-315	F	H, C1, C2	-	100	310	270	3,5	4,6	0,056
100-315	K	H, C1, C2	2	80	312	254	4	5,2	0,15
100-316	K	H, C1, C2	2	76	309	235	4,9	6,4	0,15
100-400	K	C1, C2	2	76	408	355	9,2	12	1,1
100-401	F	H, C1, C2	-	100	390	325	7,6	9,8	0,248
100-401	K	H, C1, C2	2	50	404	310	9,3	12,1	0,504
150-315	F	H, C1, C2	-	120	290	250	1,8	2,3	0,144
150-317	K	H, C1, C2	2	76	309	250	5	6,5	0,28
150-400	K	H, C1, C2	3	76	404	300	8,4	11	0,83
150-401	F	H, C1, C2	-	135	390	270	4,2	5,5	0,248
150-403	K	H, C1, C2	2	76	408	340	8,5	11,1	0,691
150-503	K	C1, C2	2	76	508	400	10	13	0,91
151-403	K	H, C1, C2	2	76	408	340	9,3	11,9	0,691
200-315	K	H, C1, C2	3	70	295	245	1,9	2,4	0,22
200-316	K	H, C1, C2	2	100	305	265	1,7	2,2	0,22
200-402	K	H, C1, C2	3	80	408	300	6,5	9,8	0,84
200-403	K	H, C1, C2	2	90	408	300	6,5	8,5	0,931
200-504	K	C1, C2	3	76	508	400	9,7	12,6	0,83
200-505	K	C1, C2	2	90	508	400	9,8	12,8	1,636
250-401	K	H, C1, C2	2	105	400	310	6	7,8	0,55
250-403	K	H, C1, C2	2	107	408	300	7	9,1	1,13
300-400	K	H, C1, C2	3	100	408	332	3,5	4,6	0,75
300-401	K	H, C1, C2	2	135	408	367	2,3	2,9	0,75
300-403	K	H, C1, C2	2	102	408	300	3,8	5	1,439
300-420	K	C1, C2	3	100	408	370	5,6	7,3	0,95
300-500	K	C1, C2	3	90	504	430	6,2	8	1,48
350-500	K	C1, C2	3	110	508	426	5,7	7,4	3,12
350-503	K	C1, C2	2	140	508	400	4,6	6	4,073
350-632	K	C1, C2	3	140	638	500	6,5	8,4	6,451
350-633	K	C1, C2	2	135	638	500	9,4	12,2	6,979
350-710	K	C1, C2	3	110	730	580	9,4	12,2	10,6
350-713	K	C1, C2	2	125	738	580	12,2	16	14,557
400-500	K	C1, C2	3	130	508	443	3,4	4,5	3,37
400-632	K	C1, C2	3	142	638	527	5,7	7,41	9,074
401-713	K	C1, C2	2	143	738	580	11,5	15	15,894
500-634	K	C1, C2	3	133	638	448	4,5	6	10,248
600-520	K	C1, C2	3	145	532	457	2,4	3,2	7,02
600-710	K	C1, C2	3	165	736	685	4,2	5,5	16,96
700-901	K	C1, C2	3	180	908	760	7,2	9,3	50
700-902	K	C1, C2	3	190	887	680	3,5	4,6	40

Moments of inertia depending on the motor

Table 14: 2 poles

Motor	Motor type	J [kgm ²]
3 2 E	1	0,002
4 2 E	1	0,005
5 2 E	1	0,006
7 2 E	1	0,011
11 2 E	2	0,012
15 2 E	2	0,022
18 2 E	2	0,034
22 2 E	2	0,044
26 2 E	2	0,044
30 2 E	3	0,06
37 2 E	3	0,08
55 2 E	3	0,190
65 2 E	3	0,220
75 2 E	3	0,250
18 2 F	5	0,034
22 2 F	5	0,044
26 2 F	5	0,044
30 2 F	5	0,077
37 2 F	5	0,097
55 2 F	5	0,189
65 2 F	5	0,219
75 2 F	5	0,249

Table 15: 4 poles

Motor	Motor type	J [kgm ²]
2 4 E	1	0,003
3 4 E	1	0,004
4 4 E	1	0,011
5 4 E	1	0,011
7 4 E	2	0,022
11 4 E	2	0,032
15 4 E	2	0,054
18 4 E	2	0,064
22 4 E	2	0,074
30 4 E	3	0,136
37 4 E	3	0,176
45 4 E	3	0,263
55 4 E	3	0,323
65 4 E	3	0,380
75 4 E	3	0,450
15 4 F	5	0,054
18 4 F	5	0,064
22 4 F	5	0,074
30 4 F	5	0,137
37 4 F	5	0,177
55 4 F	5	0,326
65 4 F	5	0,379
75 4 F	5	0,449
35 4 N	4	0,25
50 4 N	4	0,28
65 4 N	4	0,33
80 4 N	4	0,46
95 4 N	4	0,55
110 4 N	4	0,63
130 4 N	4	1,26

Motor	Motor type	J [kgm ²]
155 4 N	4	1,43
175 4 N	4	1,57
200 4 N	4	3,78
250 4 N	4	4,13
300 4 N	4	4,82
350 4 N	4	5,51

Table 16: 6 poles

Motor	Motor type	J [kgm ²]
7 6 E	2	0,032
11 6 E	2	0,042
15 6 E	2	0,094
18 6 E	2	0,114
22 6 E	3	0,186
30 6 E	3	0,216
31 6 E	3	0,463
37 6 E	3	0,463
45 6 E	3	0,550
55 6 E	3	0,650
15 6 F	5	0,094
18 6 F	5	0,114
22 6 F	5	0,187
30 6 F	5	0,217
31 6 F	5	0,466
37 6 F	5	0,466
45 6 F	5	0,549
55 6 F	5	0,649
32 6 N	4	0,37
40 6 N	4	0,45
50 6 N	4	0,54
60 6 N	4	0,66
80 6 N	4	0,80
100 6 N	4	0,94
120 6 N	4	1,98
140 6 N	4	2,25
165 6 N	4	2,55
190 6 N	4	7,30
225 6 N	4	8,57
260 6 N	4	9,84
320 6 N	4	14,32
360 6 N	4	15,89
400 6 N	4	17,58
440 6 N	4	19,15
480 6 N	4	20,71
530 6 N	4	32,54
580 6 N	4	37,34
630 6 N	4	42,06
690 6 N	4	46,82
770 6 N	4	51,58
850 6 N	4	56,34

Table 17: 8 poles

Motor	Motor type	J [kgm ²]
11 8 E	3	0,186
15 8 E	3	0,186
18 8 E	3	0,226
22 8 E	3	0,276

Motor	Motor type	J [kgm ²]
30 8 E	3	0,463
37 8 E	3	0,550
45 8 E	3	0,650
15 8 F	5	0,187
18 8 F	5	0,227
22 8 F	5	0,277
30 8 F	5	0,466
37 8 F	5	0,549
45 8 F	5	0,649
26 8 N	4	0,40
35 8 N	4	0,50
50 8 N	4	0,66
65 8 N	4	0,80
75 8 N	4	0,94
90 8 N	4	1,98
110 8 N	4	2,25
130 8 N	4	2,55
150 8 N	4	7,30
185 8 N	4	8,57
220 8 N	4	9,84
260 8 N	4	13,27
300 8 N	4	15,88
350 8 N	4	19,13
400 8 N	4	20,70
460 8 N	4	32,54
530 8 N	4	37,30
580 8 N	4	42,06
630 8 N	4	46,82
690 8 N	4	51,58
760 8 N	4	56,34

Table 18: 10 poles

Motor	Motor type	J [kgm ²]
40 10 N	4	1,75
60 10 N	4	1,93
75 10 N	4	2,20
90 10 N	4	2,49
110 10 N	4	7,96
150 10 N	4	9,66
190 10 N	4	11,83
230 10 N	4	17,73
270 10 N	4	20,50
310 10 N	4	23,15
350 10 N	4	25,79
390 10 N	4	37,12
430 10 N	4	42,64
475 10 N	4	48,17
535 10 N	4	53,69
600 10 N	4	59,21
660 10 N	4	64,73

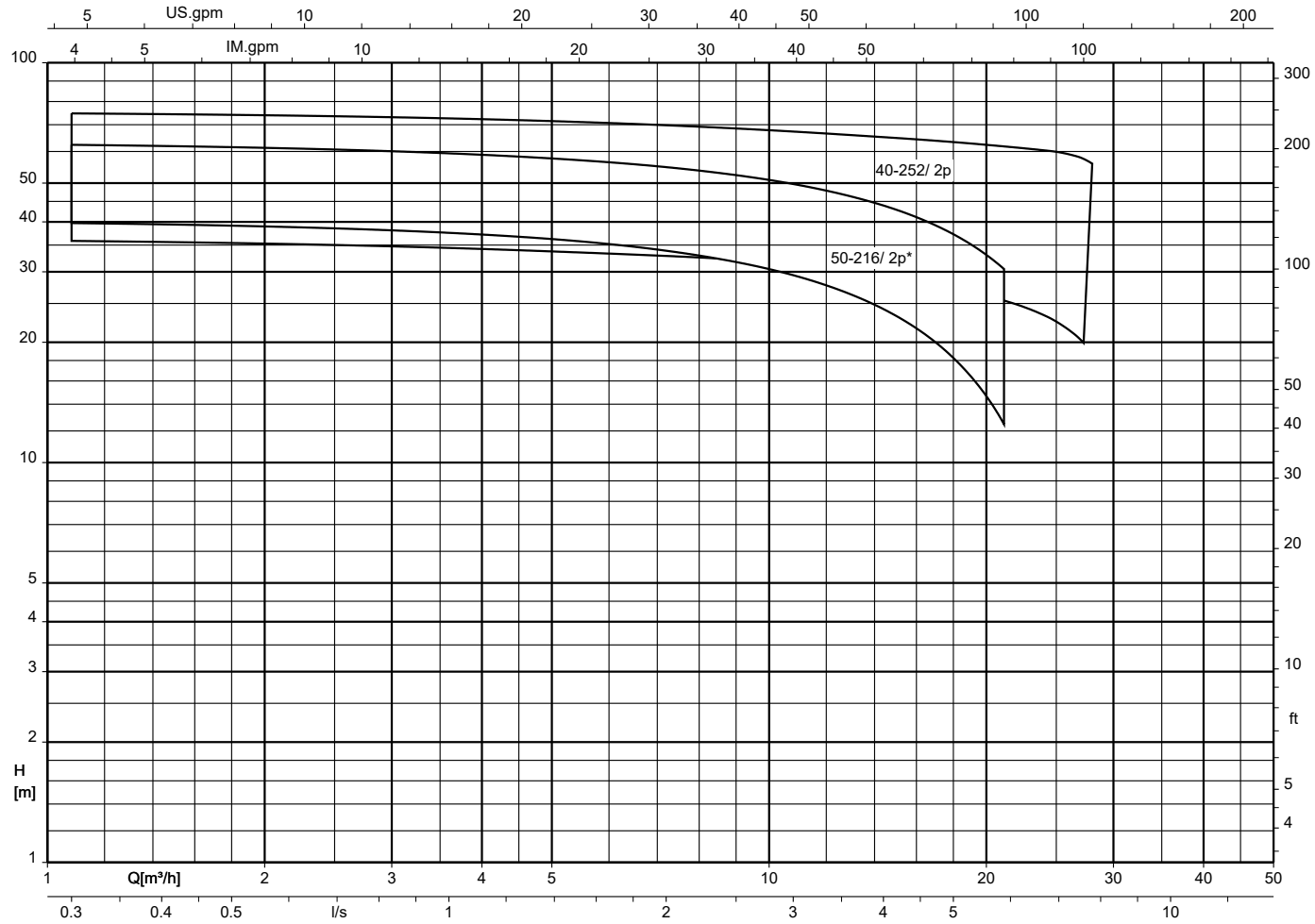
Table 19: 12 poles

Motor	Motor type	J [kgm ²]
195 12 N	4	17,73
230 12 N	4	20,50
265 12 N	4	23,15
300 12 N	4	25,79

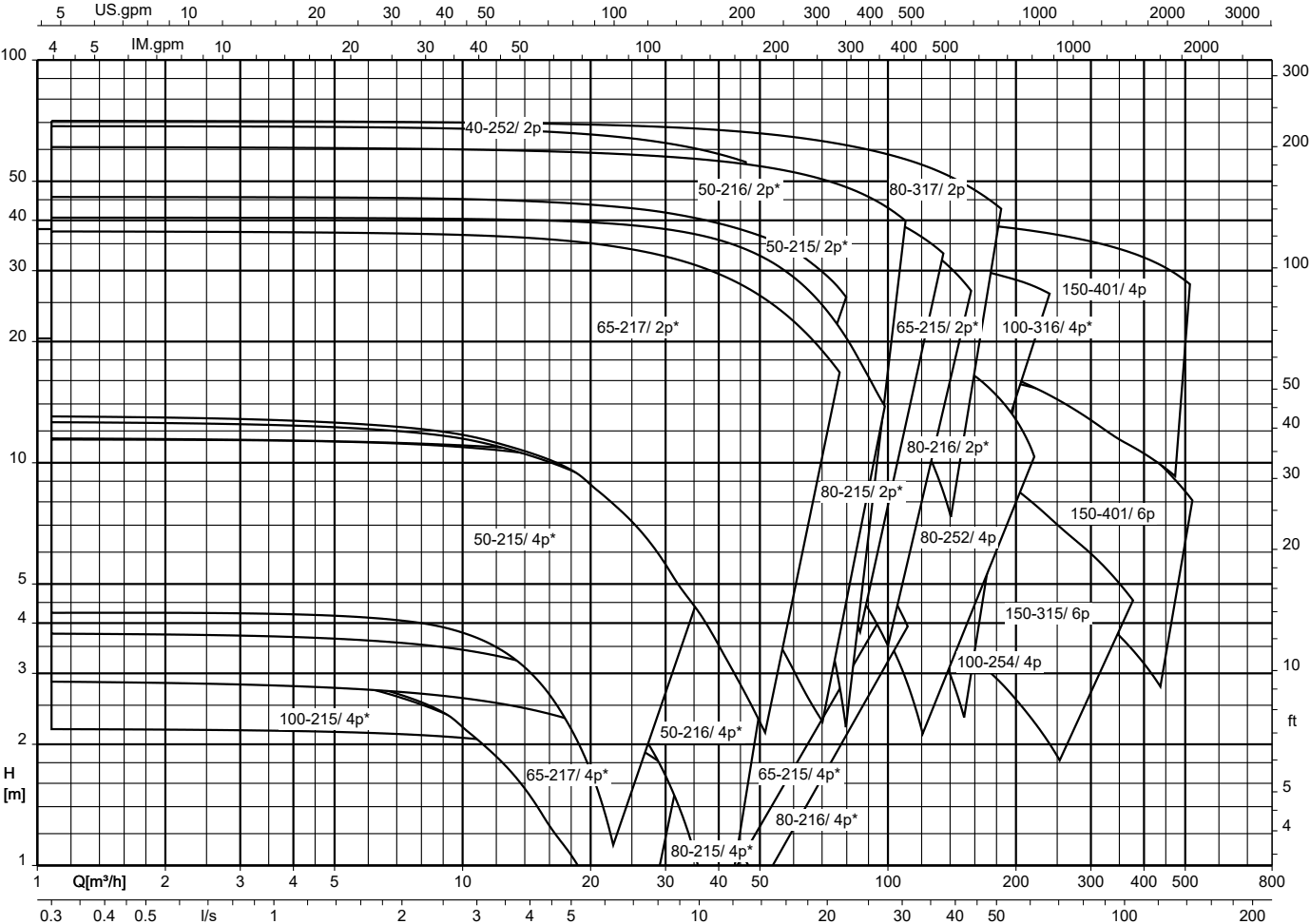
Motor	Motor type	J [kgm ²]
340 12 N	4	42,64
380 12 N	4	48,17
450 12 N	4	53,69
490 12 N	4	59,21
560 12 N	4	64,73

Selection charts

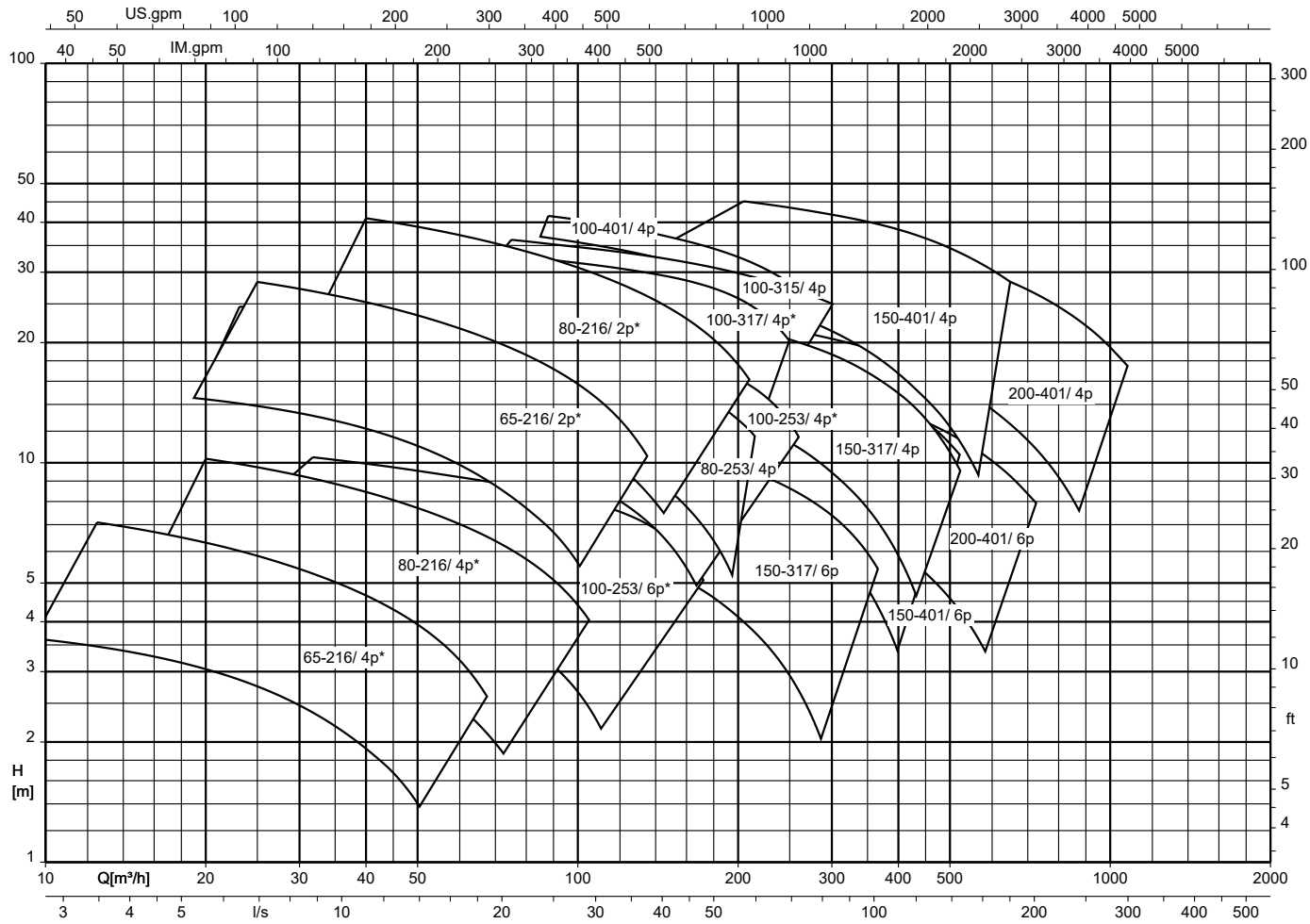
Amarex KRT S + *S-max, n = 2900 rpm



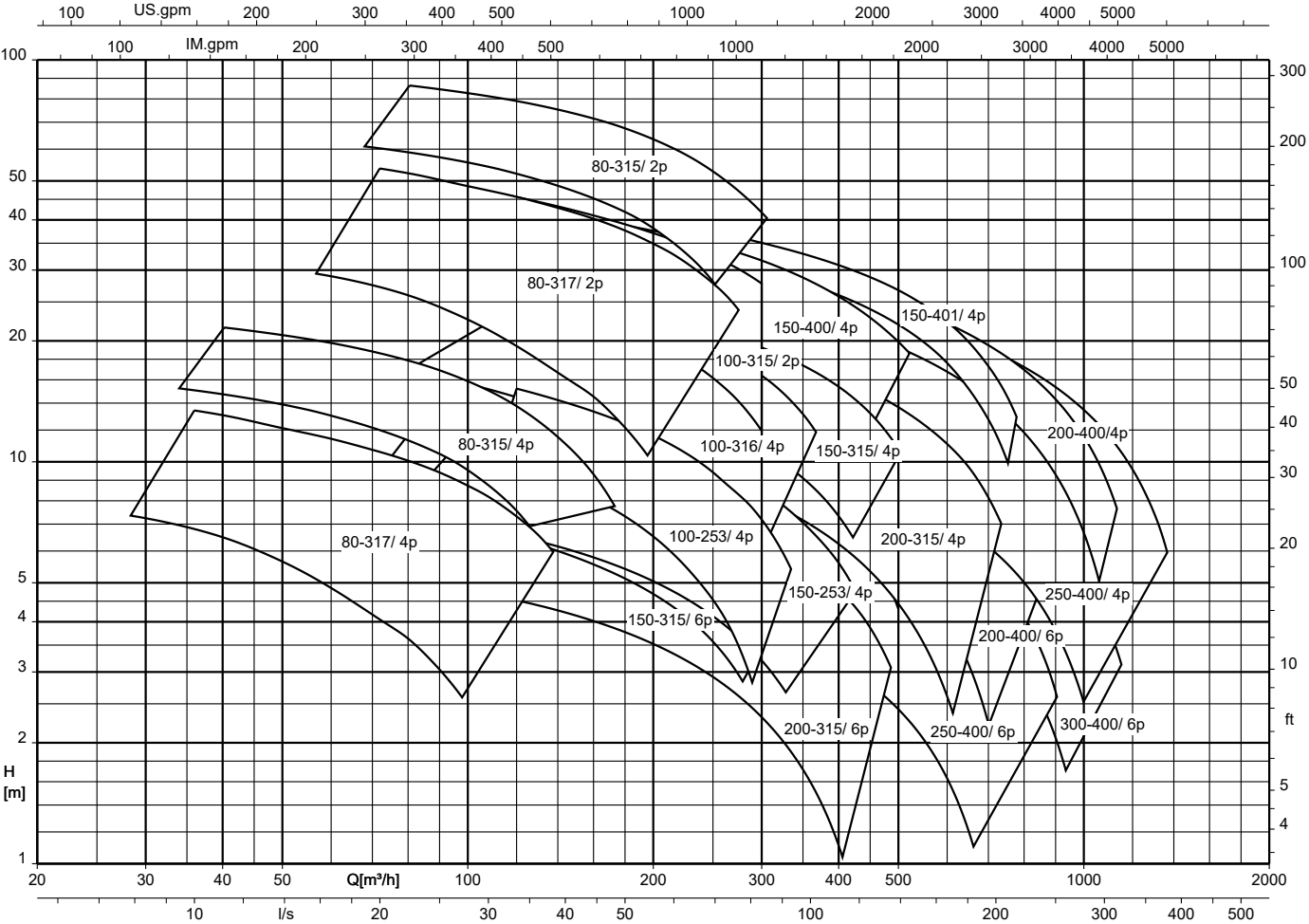
Amarex KRT F + *F-max, n = 2900/1450/960 rpm



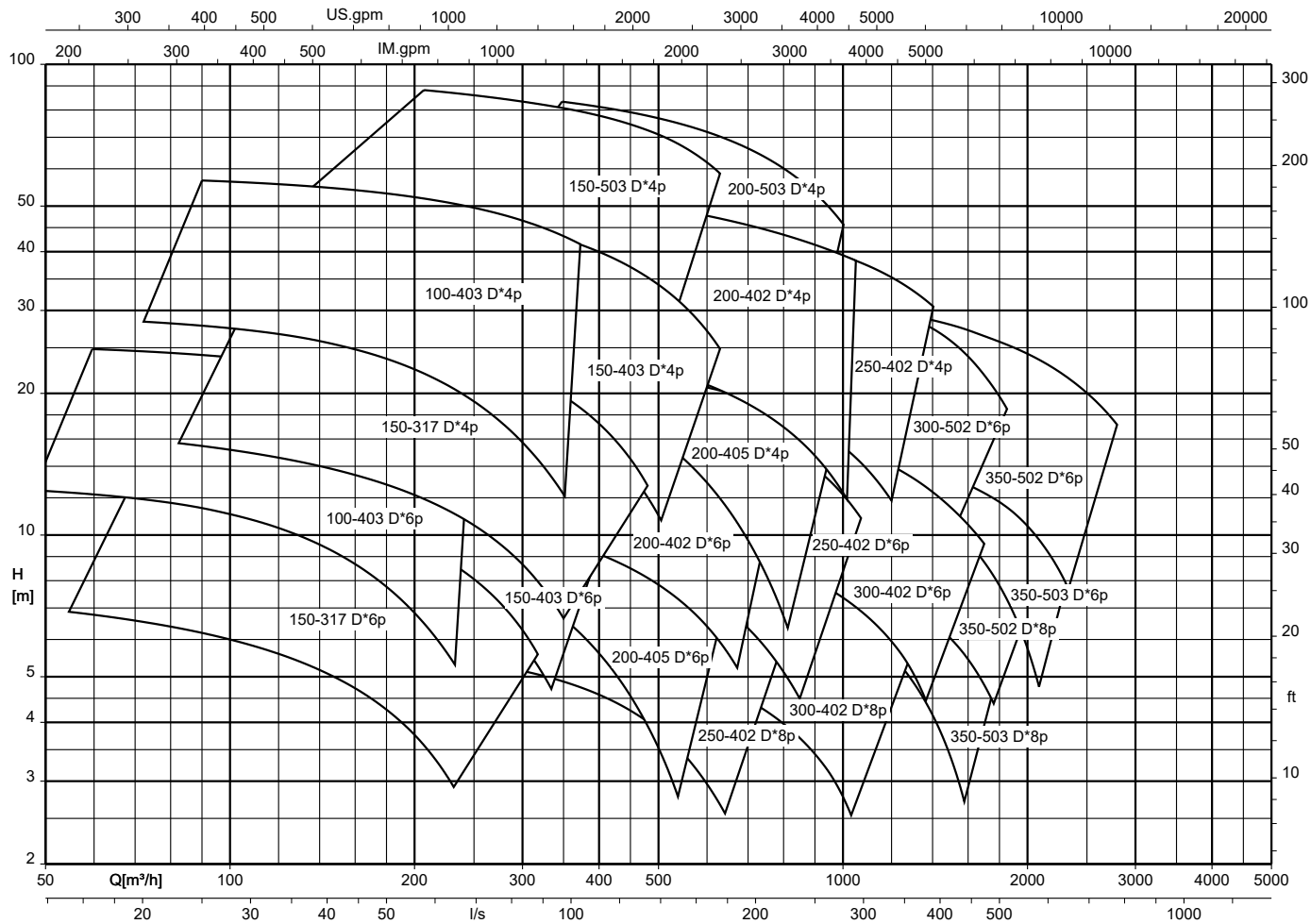
Amarex KRT E + *E-max, n = 2900/1450/960 rpm



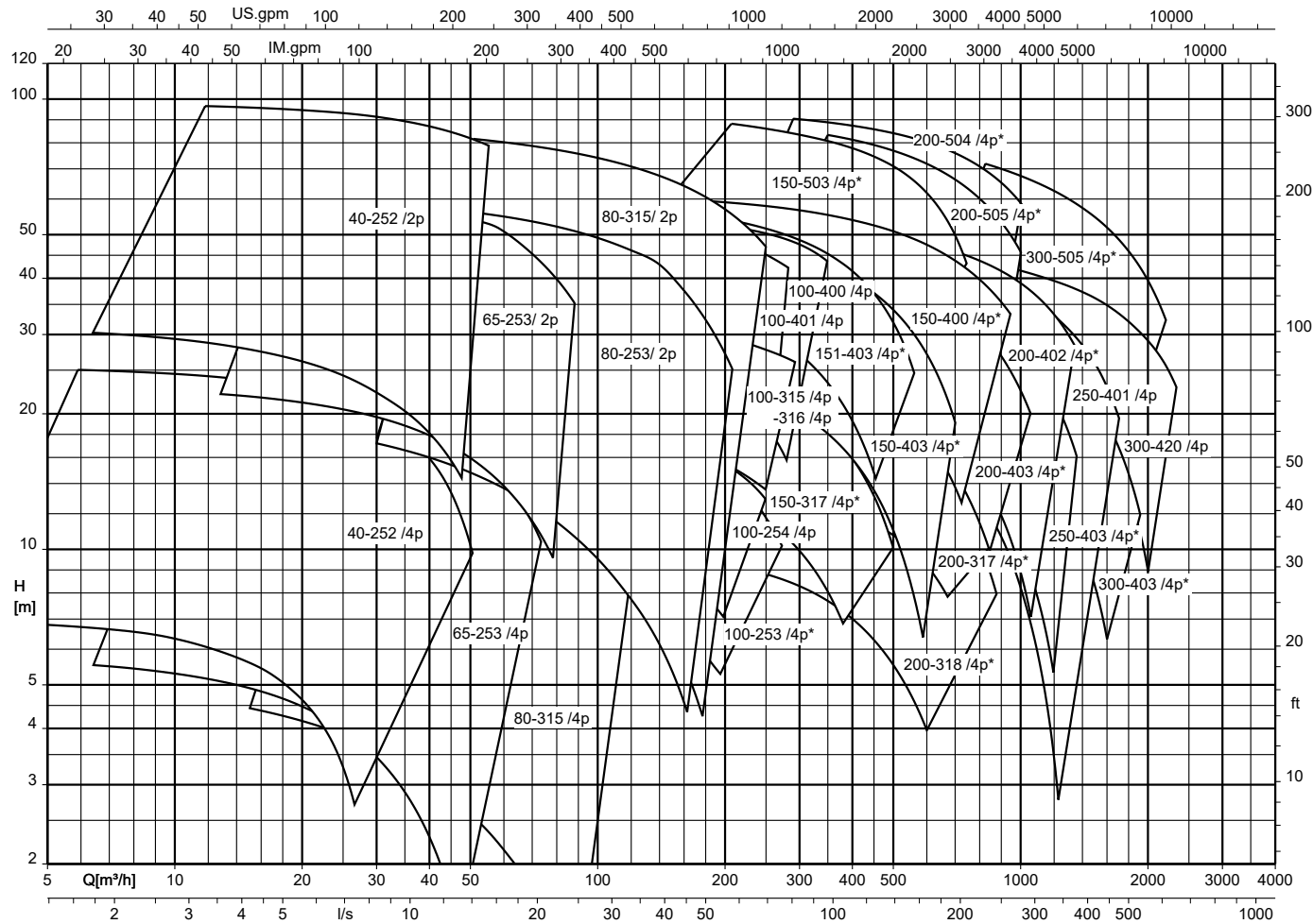
Amarex KRT D, n = 2900/1450/960 rpm



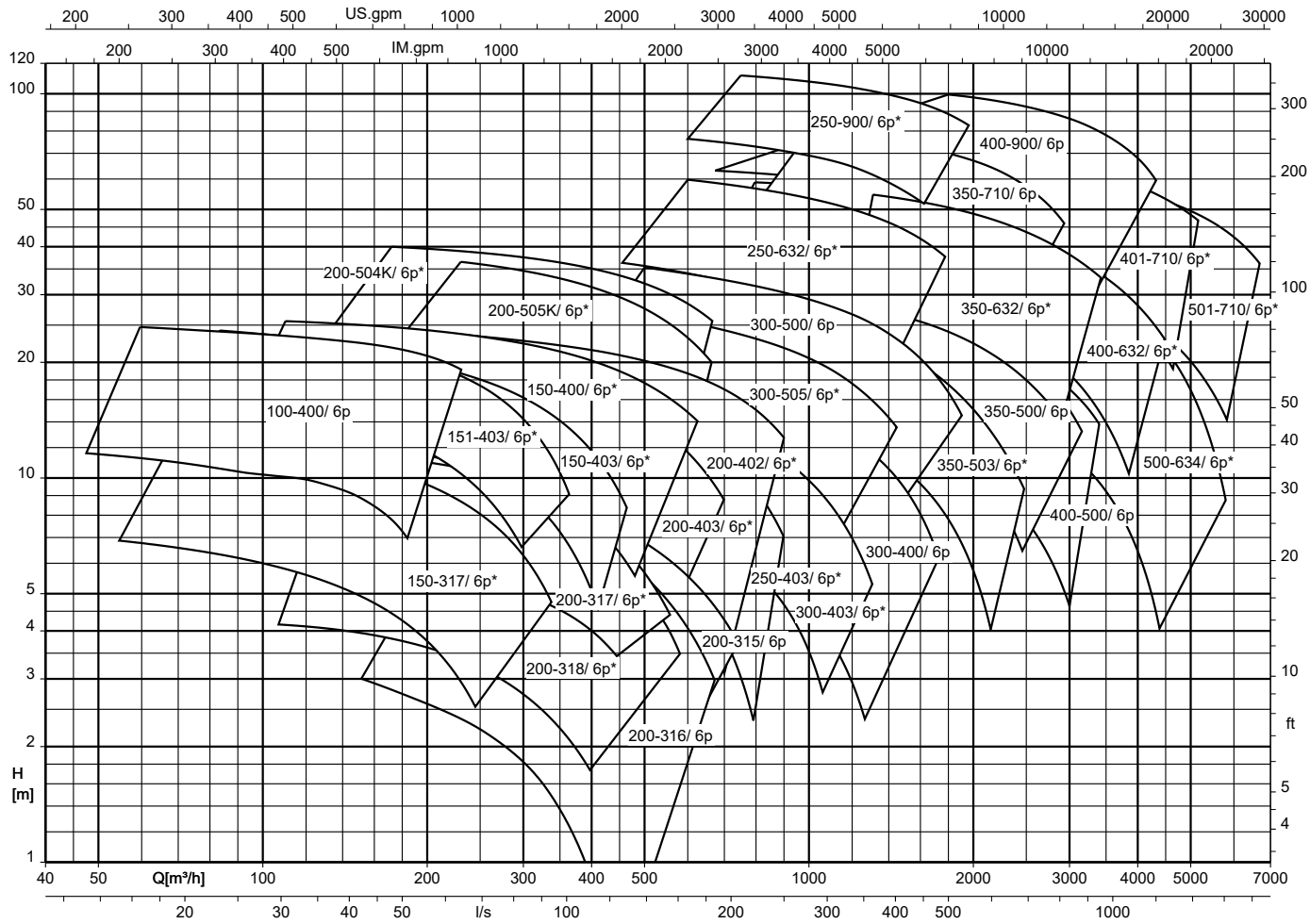
Amarex KRT D-max, n = 1450/960/725 rpm



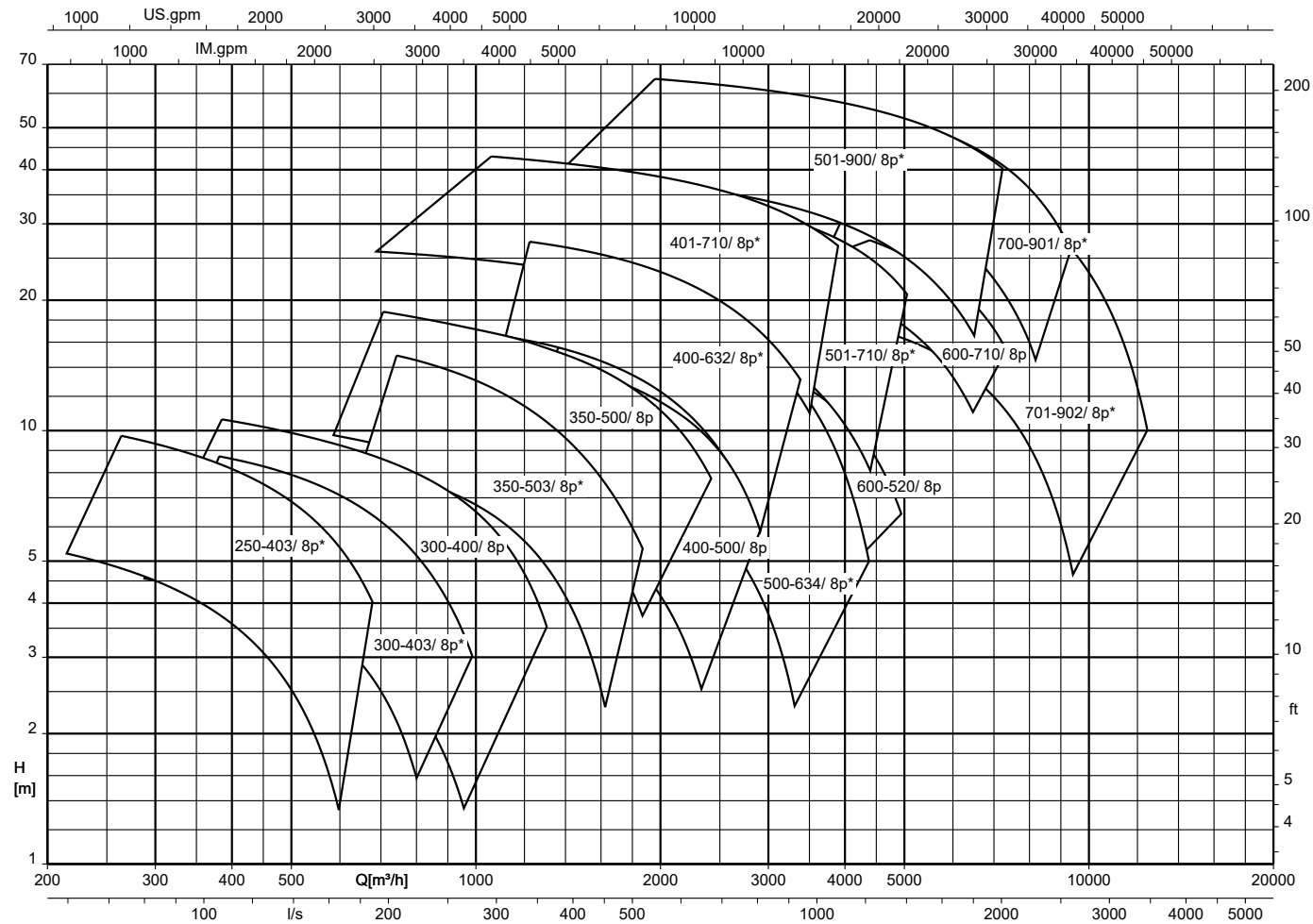
Amarex KRT K + *K-max, n = 2900/1450 rpm



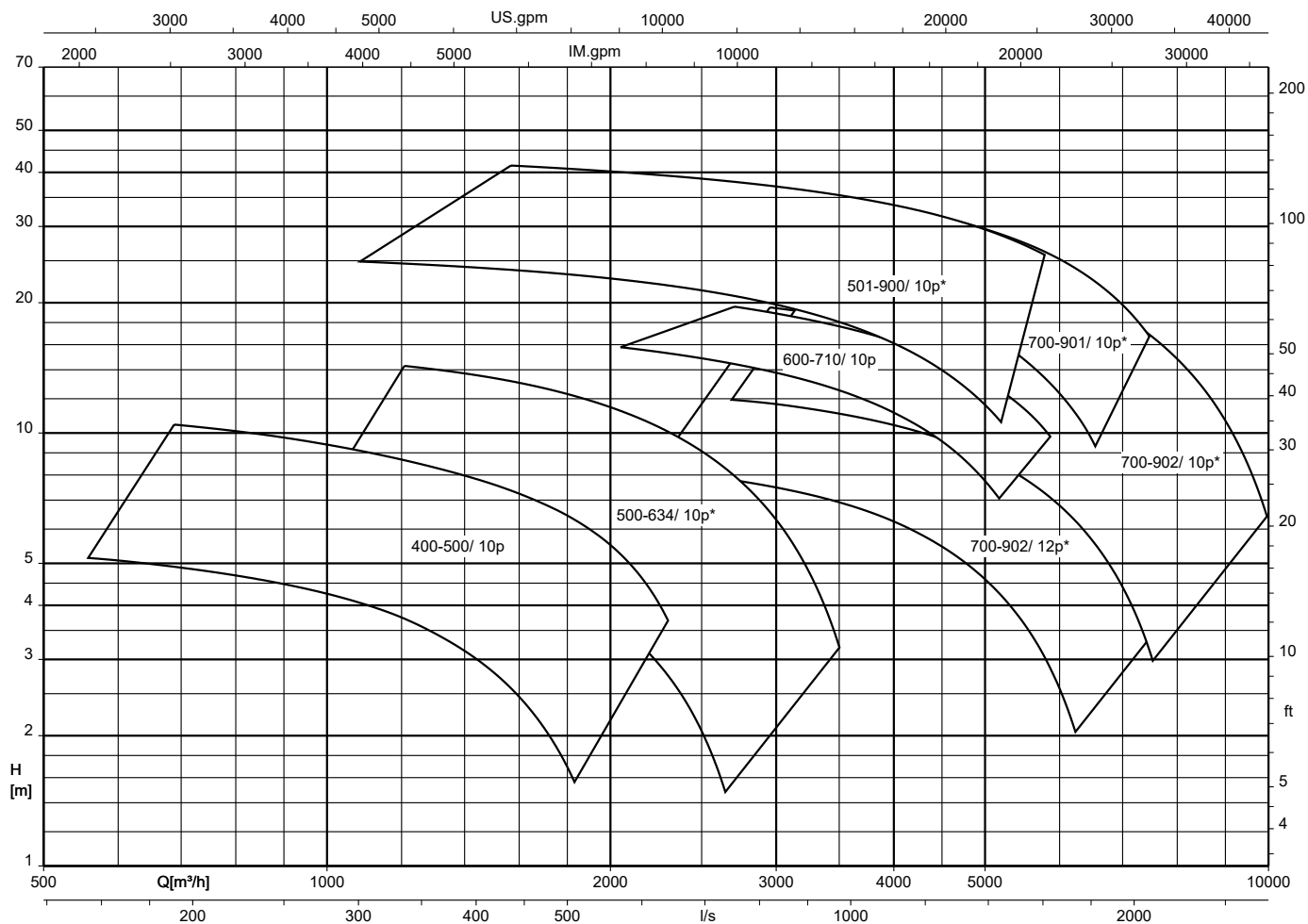
Amarex KRT K + *K-max, n = 960 rpm



Amarex KRT K + *K-max, n = 725 rpm



Amarex KRT K + *K-max, n = 580/480 rpm



Installation types

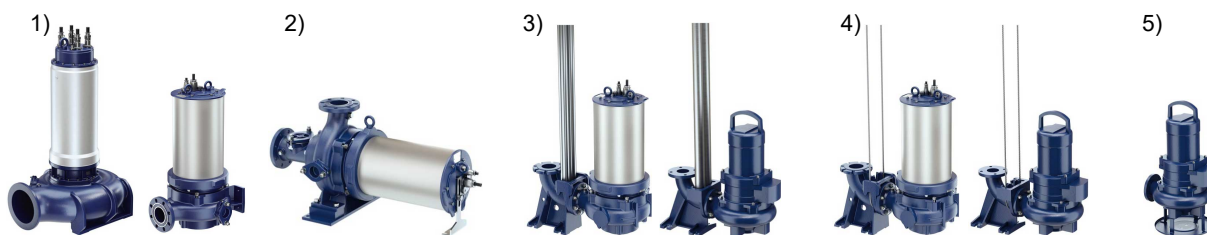


Fig. 1: Installation types

1	Installation type D: stationary dry installation, vertical (S1 duty)
2	Installation type H: stationary dry installation, horizontal (S1 duty)
3	Installation type K: stationary wet installation (S1 duty with motor outside of the fluid possible) with guide rail arrangement Installation type S: stationary wet installation (S1 duty with submerged motor) with guide rail arrangement
4	Installation type K: stationary wet installation (S1 duty with motor outside of the fluid possible) with guide wire arrangement Installation type S: stationary wet installation (S1 duty with submerged motor) with guide wire arrangement
5	Installation type P: wet installation of transportable model (S1 duty with submerged motor)

Pump sets of installation types D, H and K

are suitable for continuous duty with the motor outside the fluid. Cooling is effected by means of air convection. Versions with a cooling jacket have an additional internal cooling circuit.

Pump sets of installation types P and S

are designed for continuously submerged operation. The motor is cooled by the fluid handled on the motor surface. Operation with the motor outside the fluid handled is possible for short periods.

Scope of supply

Stationary dry installation - vertical (installation type D)

- Pump set complete with electric cables
- Suction elbow with inspection hole³⁸⁾ (optional) and fastening elements
- Optional: suction elbow with inspection hole

Stationary dry installation - horizontal (installation type H)

- Pump set complete with electric cables
- Foundation rails³⁹⁾
- Suction-side flanged spacer with inspection hole⁴⁰⁾ (optional)
- Amaslide maintenance aid (optional)

Stationary wet installation (installation types K and S)

- Pump set complete with electric cables
- Claw with sealing elements and fasteners
- Lifting rope, lifting chain or lifting bail (optional)
- Mounting bracket with fasteners
- Flanged bend and fastening elements
- Guide wire / guide rail (guide rails are not included in KSB's scope of supply)

Transportable wet-installed model (installation type P)

- Pump set complete with electric cables
- Foot plate or pump stool with fasteners

³⁸ For nominal discharge nozzle diameter $\geq$ DN100

³⁹ For UE/XE motors only

⁴⁰ For nominal discharge nozzle diameter $\geq$ DN 100

General arrangement drawings with list of components

Amarex KRT, motor type 1

For the motor type assigned to the motor see (⇒ Page 22)

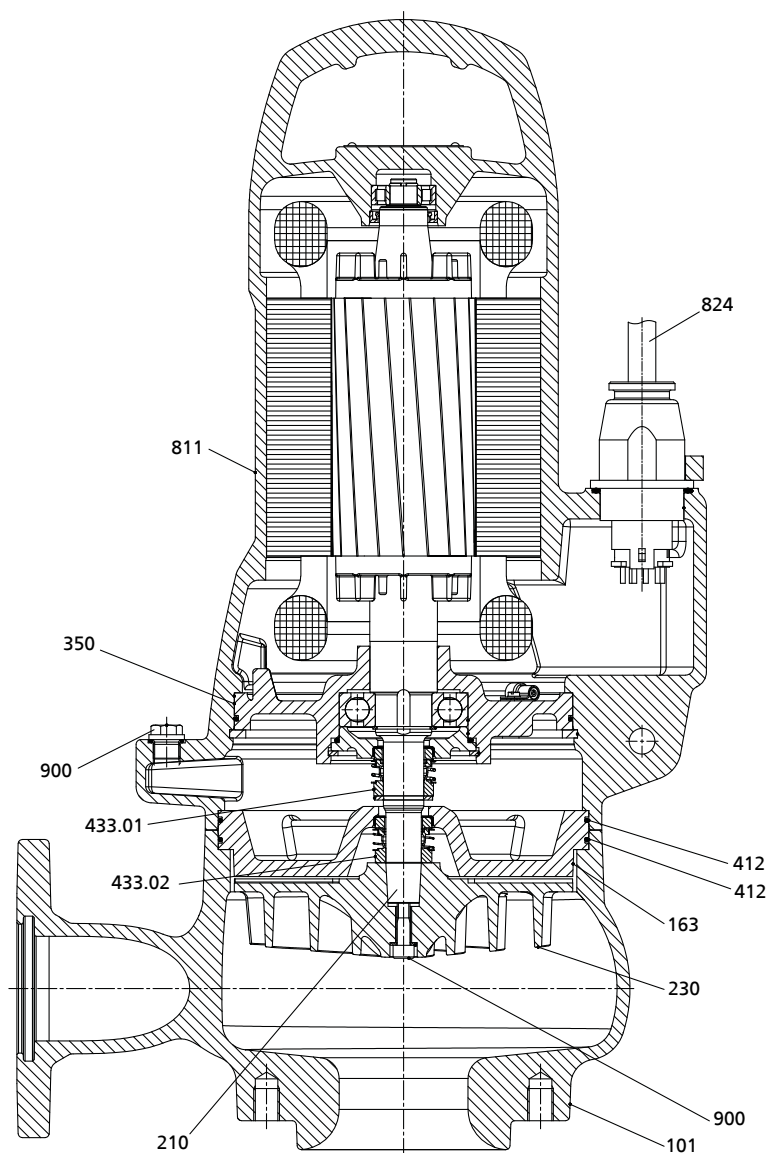


Fig. 2: General assembly drawing, example: Amarex KRT F65-215

Table 20: List of components

Part No.	Description	Part No.	Description
101	Pump casing	412	O-ring
163	Discharge cover	433.01/02	Mechanical seal
210	Shaft	811	Motor housing
230	Impeller	824	Power cable
350	Bearing housing	900	Bolt/screw

Amarex KRT, motor type 2

For the motor type assigned to the motor see (⇒ Page 22)

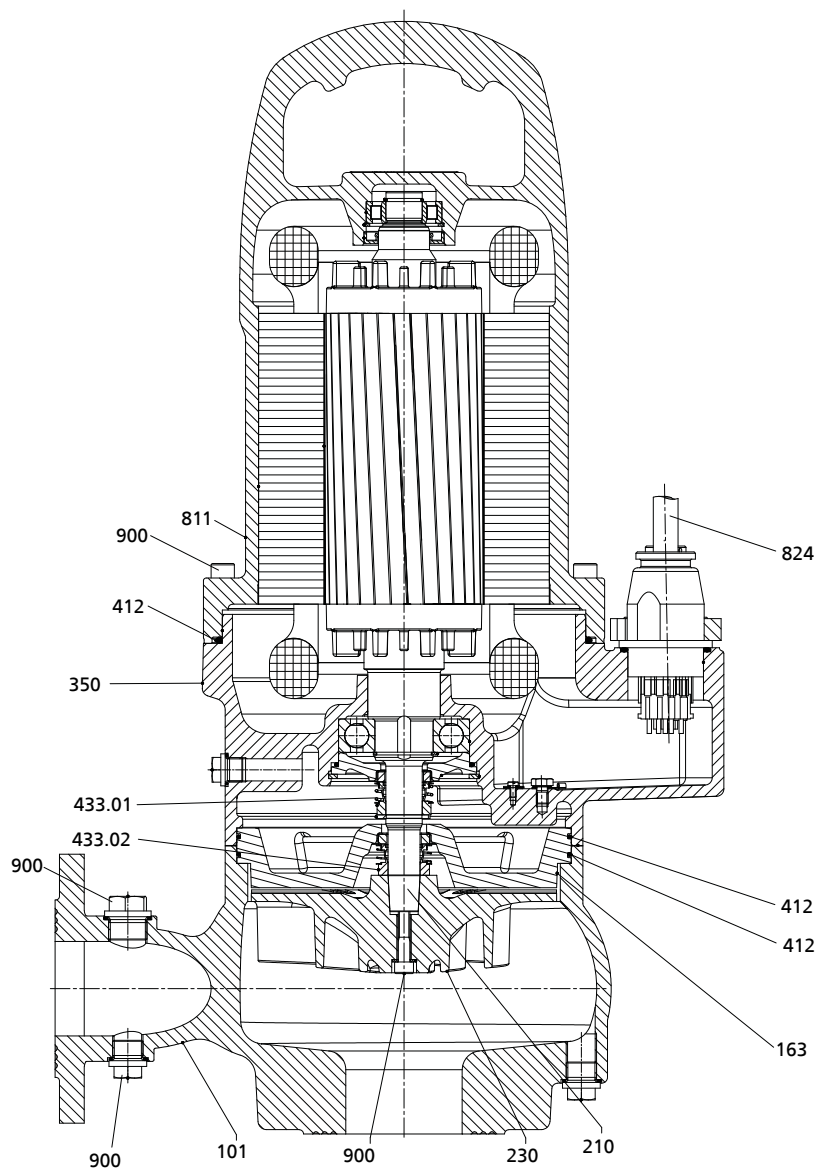


Fig. 3: General assembly drawing, example: Amarex KRT F 65-215

Table 21: List of components

Part No.	Description	Part No.	Description
101	Pump casing	412	O-ring
163	Discharge cover	433.01/02	Mechanical seal
210	Shaft	811	Motor housing
230	Impeller	824	Power cable
350	Bearing housing	900	Bolt/screw

Amarex KRT, motor type 3

For the motor type assigned to the motor see (⇒ Page 22)

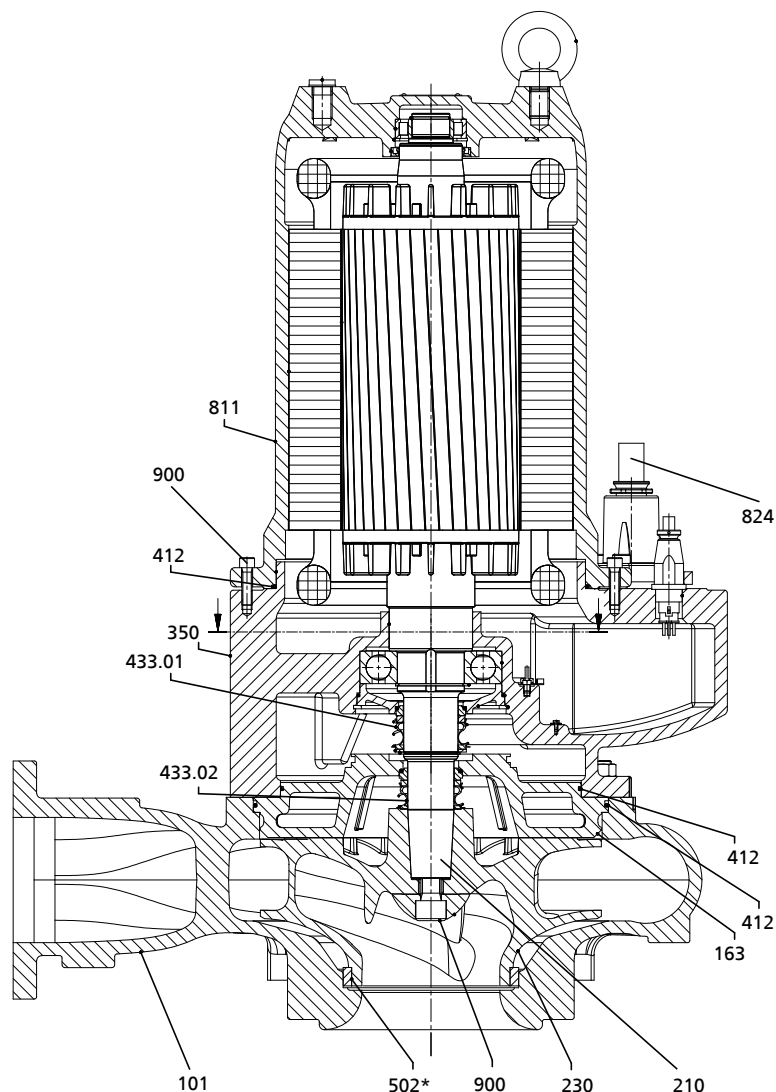


Fig. 4: General assembly drawing, example: Amarex KRT E/K 100-400/75 4 XEG

*: On specific designs only

Table 22: List of components

Part No.	Description	Part No.	Description
101	Pump casing	433.01/.02	Mechanical seal
163	Discharge cover	502	Casing wear ring
210	Shaft	811	Motor housing
230	Impeller	824	Power cable
350	Bearing housing	900	Bolt/screw
412	O-ring		

Amarex KRT, motor type 4, installation types S and P

For the motor type assigned to the motor see (⇒ Page 22)

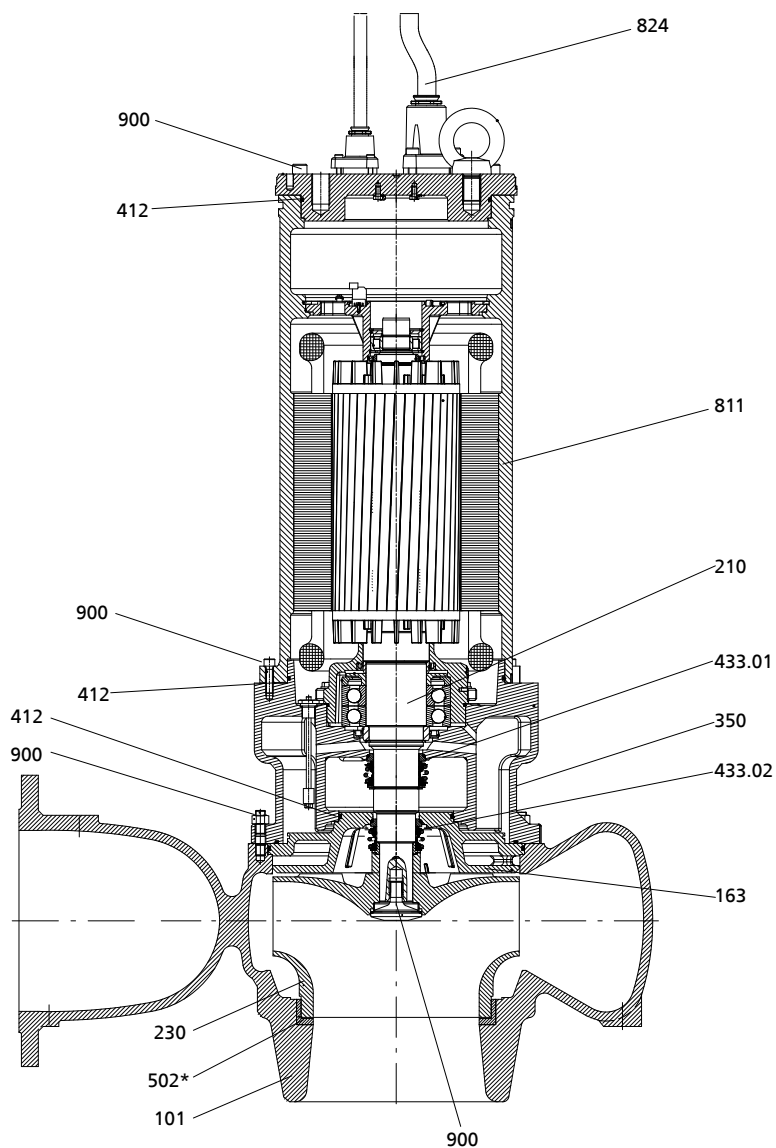


Fig. 5: General assembly drawing, example: Amarex KRT K 150-403 / 130 4 XNG-S without cooling jacket

*: On specific designs only

Table 23: List of components

Part No.	Description	Part No.	Description
101	Pump casing	433.01/.02	Mechanical seal
163	Discharge cover	502	Casing wear ring
210	Shaft	811	Motor housing
230	Impeller	824	Power cable
350	Bearing housing	900	Bolt/screw
412	O-ring		

Amarex KRT, motor type 4, installation types K and D

For the motor type assigned to the motor see (⇒ Page 22)

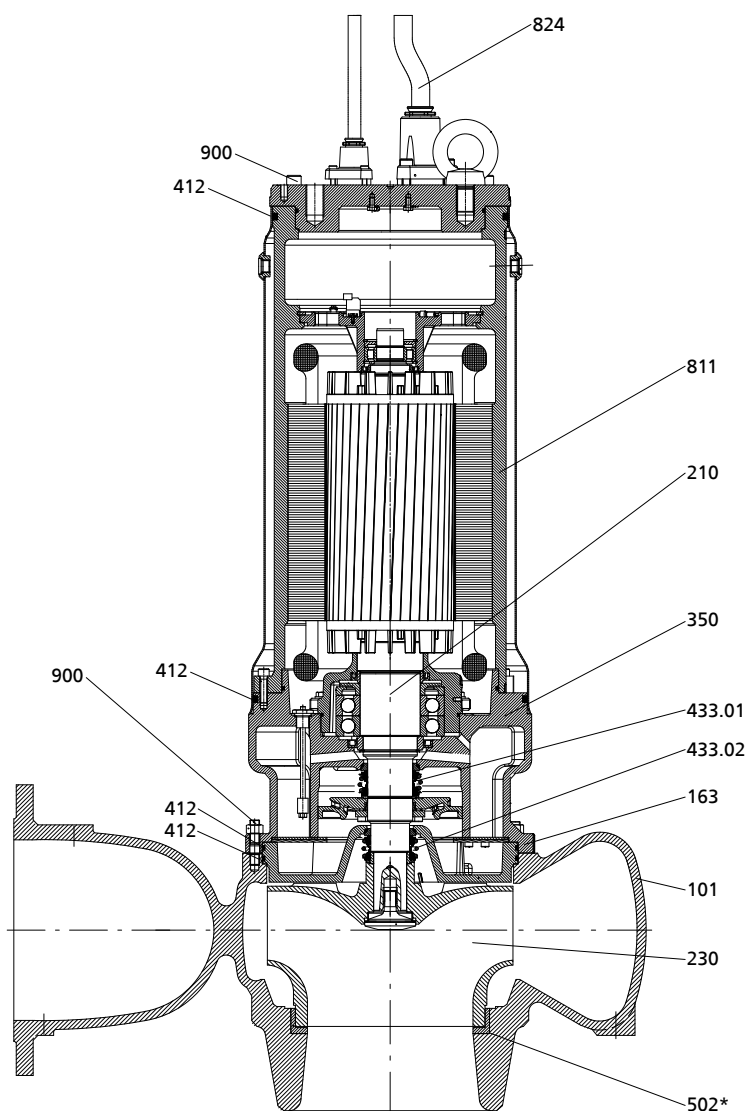


Fig. 6: General assembly drawing, example: Amarex KRT K 150-403 / 130 4 XNG-K with cooling jacket

*: On specific designs only

Table 24: List of components

Part No.	Description	Part No.	Description
101	Pump casing	433.01/.02	Mechanical seal
163	Discharge cover	502	Casing wear ring
210	Shaft	811	Motor housing
230	Impeller	824	Power cable
350	Bearing housing	900	Bolt/screw
412	O-ring		

Amarex KRT, motor type 5

For the motor type assigned to the motor see (⇒ Page 22)

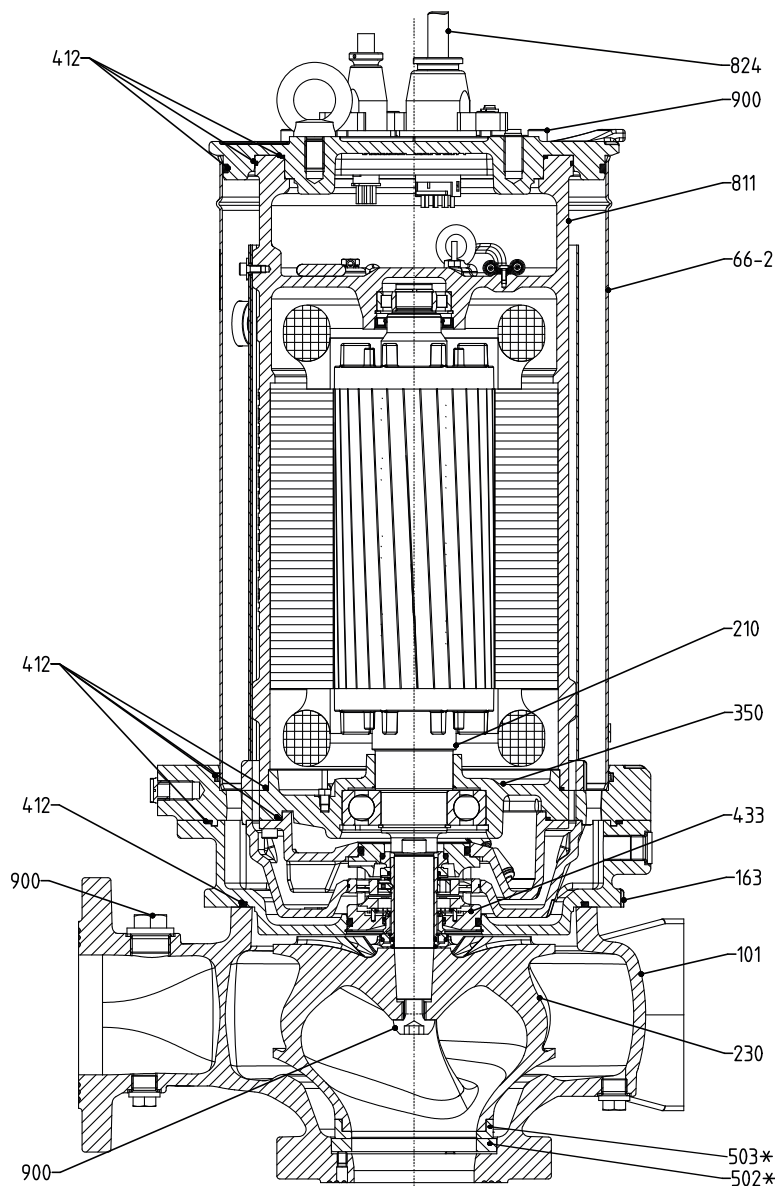


Fig. 7: General assembly drawing, example: Amarex KRT K 100-253 / 22 4 UFG-K with cooling jacket

*: On specific designs only

Table 25: List of components

Part No.	Description	Part No.	Description
101	Pump casing	502	Casing wear ring
163	Discharge cover	503	Impeller wear ring
210	Shaft	66-2	Cooling jacket
230	Impeller	811	Motor housing
350	Bearing housing	824	Power cable
412	O-ring	900	Bolt/screw
433	Mechanical seal		



LENNTECH
WATER TREATMENT SOLUTIONS

info@lennotech.com Tel. +31-152-610-900
www.lennotech.com Fax. +31-152-616-289